Digital Image Processing:

Digital Image Fundamental

What is Digital Image Processing ?

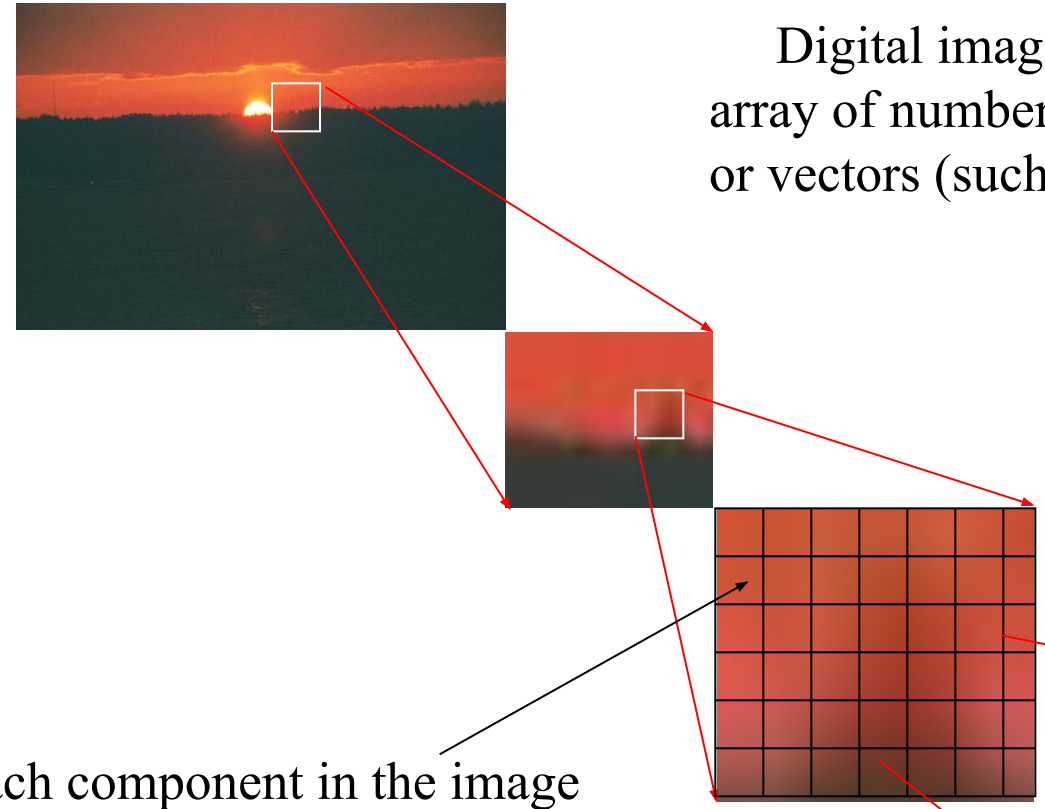
Processing of a multidimensional pictures by a digital computer

Why we need Digital Image Processing ?

Image processing focuses on two major tasks:

1. Improvement of pictorial information for human interpretation and high level processing.
2. Processing of image data for storage and transmission

Digital Image



Digital image = a multidimensional array of numbers (such as intensity image) or vectors (such as color image)

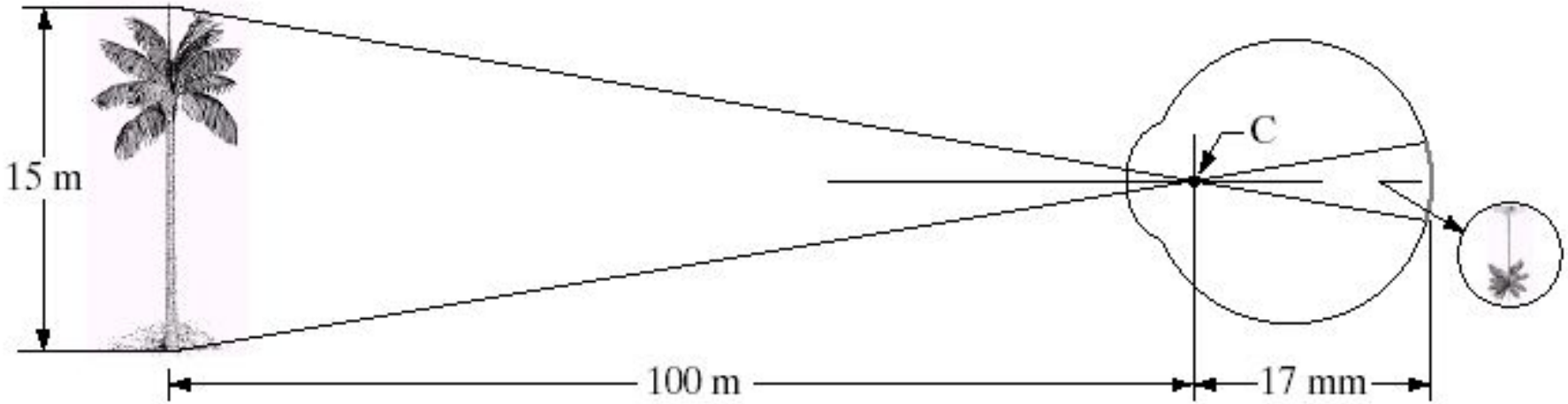
Each component in the image called pixel associates with the pixel value (a single number in the case of intensity images or a vector in the case of color images).

$$\begin{bmatrix} 10 & 10 & 16 & 28 \\ 9 & 6 & 26 & 37 \\ 15 & 25 & 13 & 22 \\ 32 & 15 & 87 & 39 \end{bmatrix} \begin{bmatrix} 65 & 70 & 56 & 43 \\ 32 & 99 & 70 & 56 \\ 21 & 54 & 90 & 96 \\ 54 & 85 & 85 & 43 \end{bmatrix} \begin{bmatrix} 78 \\ 67 \\ 42 \\ 92 \end{bmatrix}$$

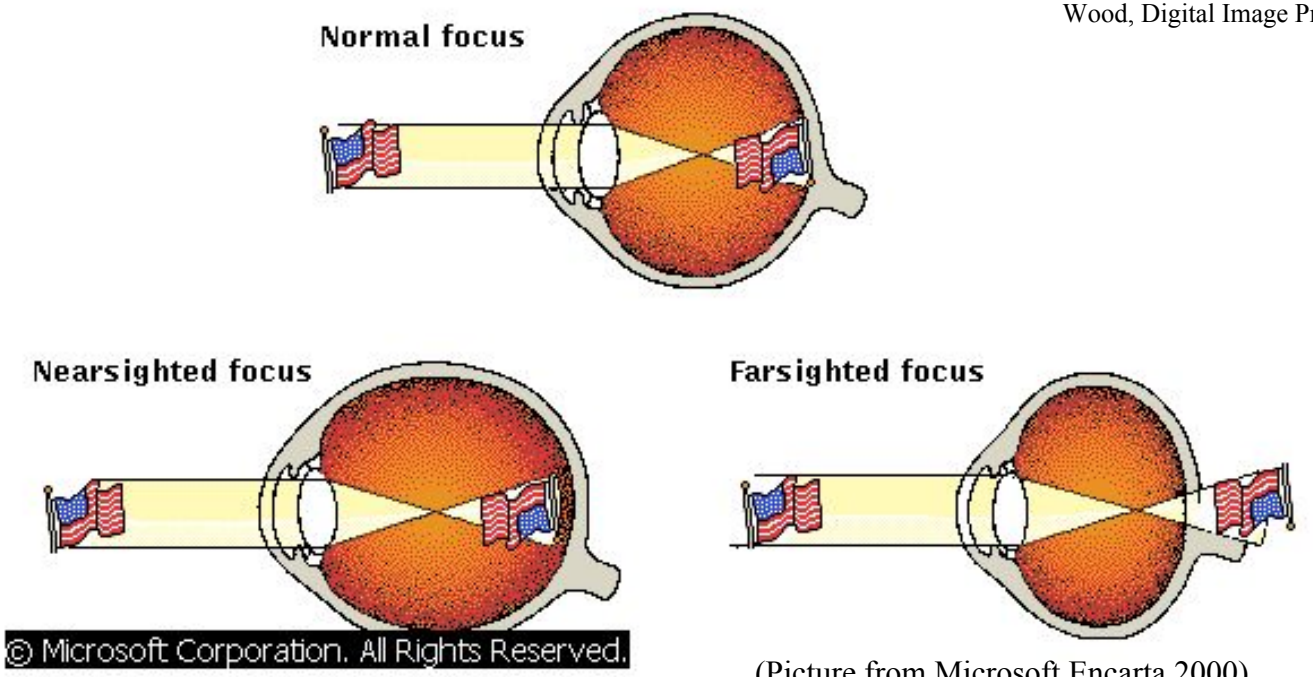
$$\begin{bmatrix} 65 & 70 & 56 & 43 \\ 32 & 54 & 96 & 67 \\ 60 & 54 & 47 & 96 \\ 85 & 65 & 65 & 39 \end{bmatrix} \begin{bmatrix} 78 \\ 67 \\ 42 \\ 92 \end{bmatrix}$$

$$\begin{bmatrix} 32 & 65 & 87 & 99 \end{bmatrix}$$

Image Formation in the Human Eye

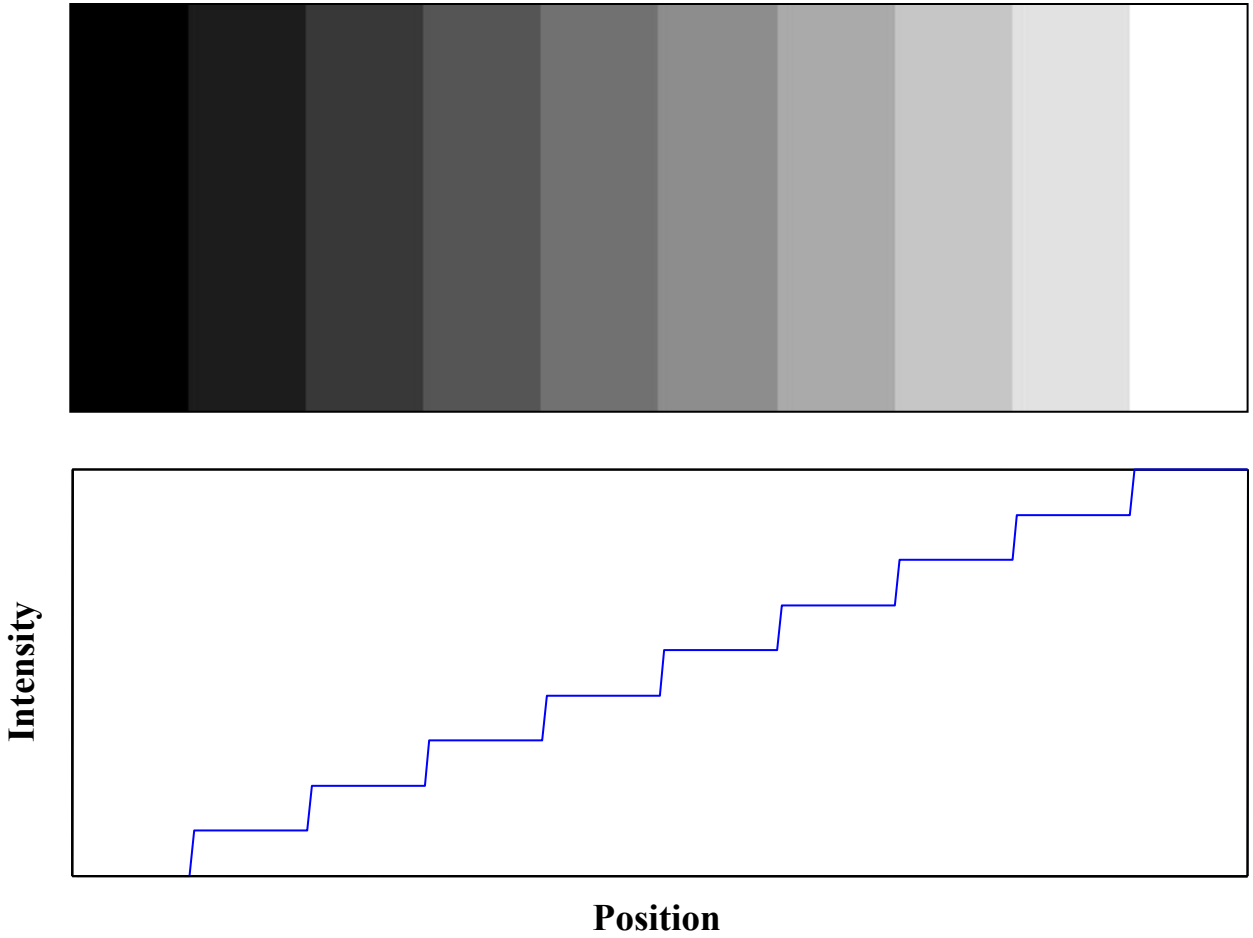


(Images from Rafael C. Gonzalez and Richard E. Wood, Digital Image Processing, 2nd Edition.

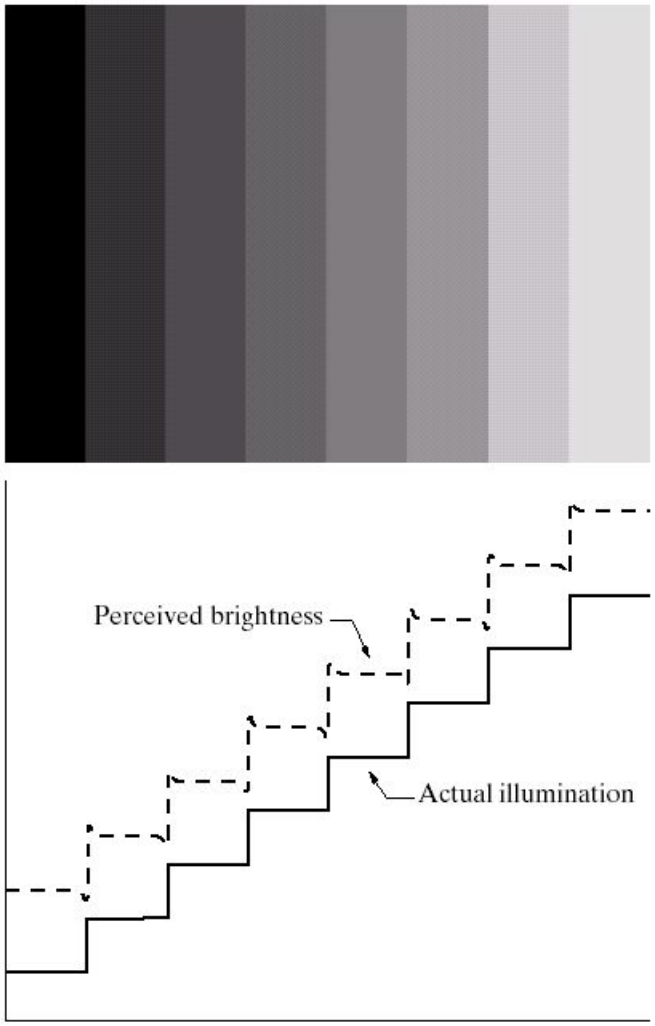


(Picture from Microsoft Encarta 2000)

Brightness Adaptation of Human Eye : Mach Band Effect



Mach Band Effect

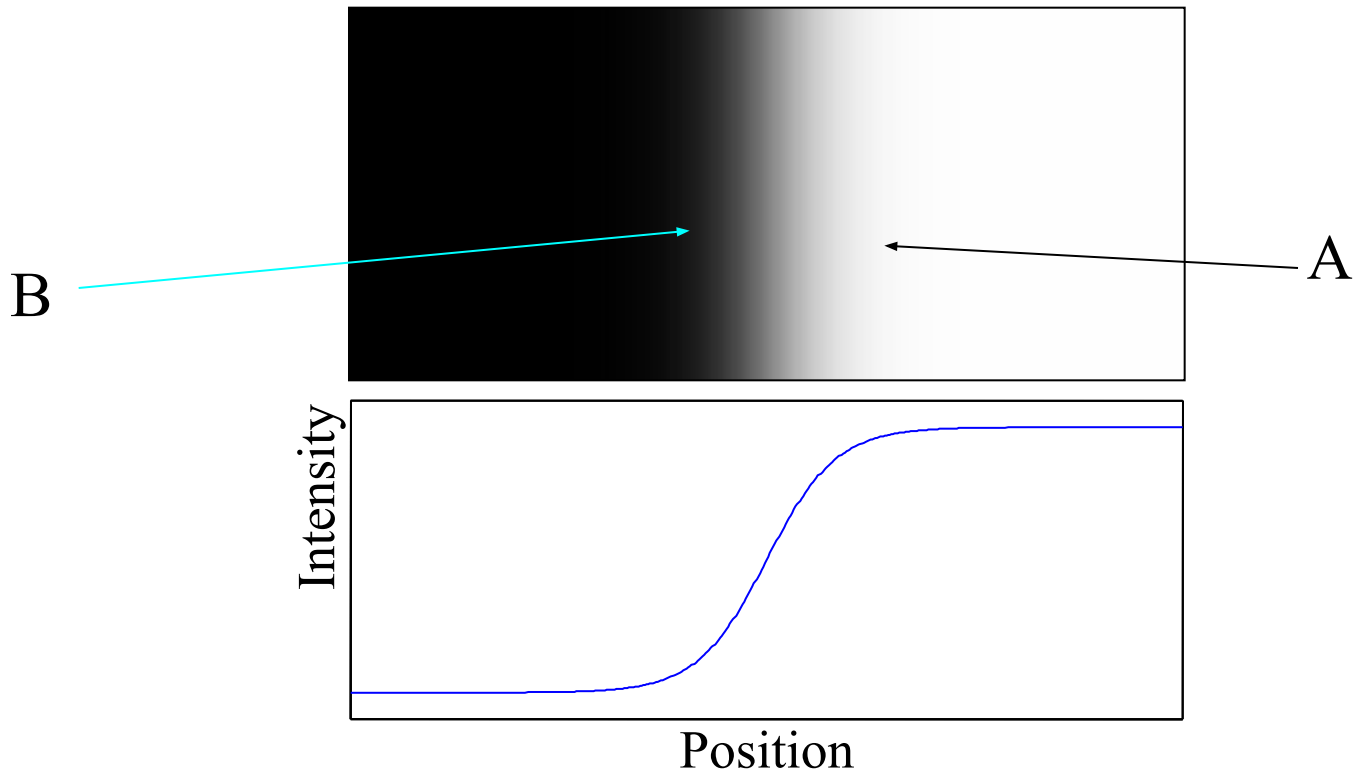


Intensities of surrounding points effect perceived brightness at each point.

In this image, edges between bars appear brighter on the right side and darker on the left side.

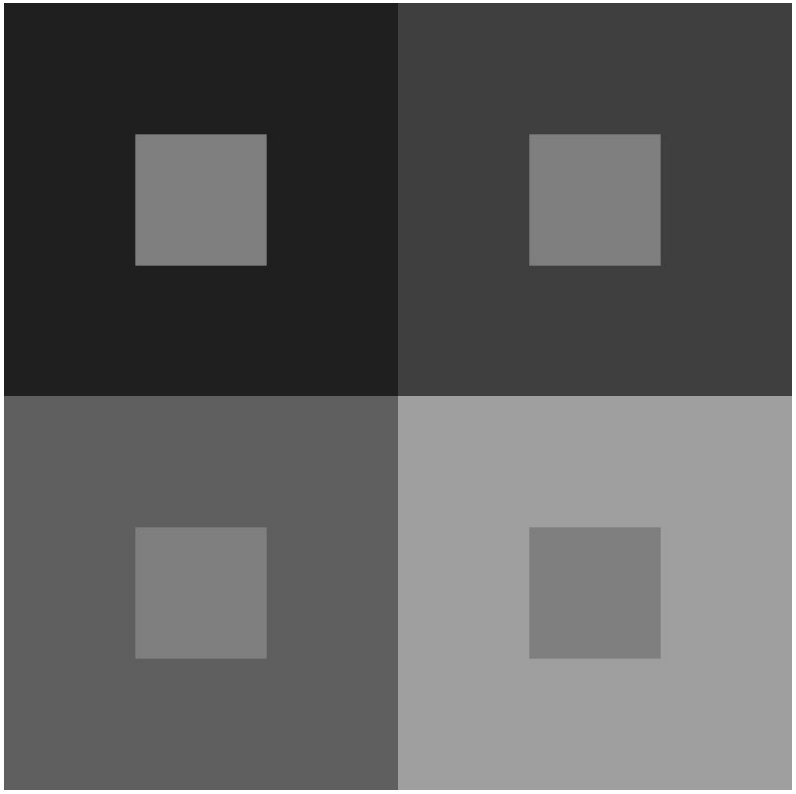
(Images from Rafael C. Gonzalez and Richard E. Wood, Digital Image Processing, 2nd Edition.

Mach Band Effect (Cont)



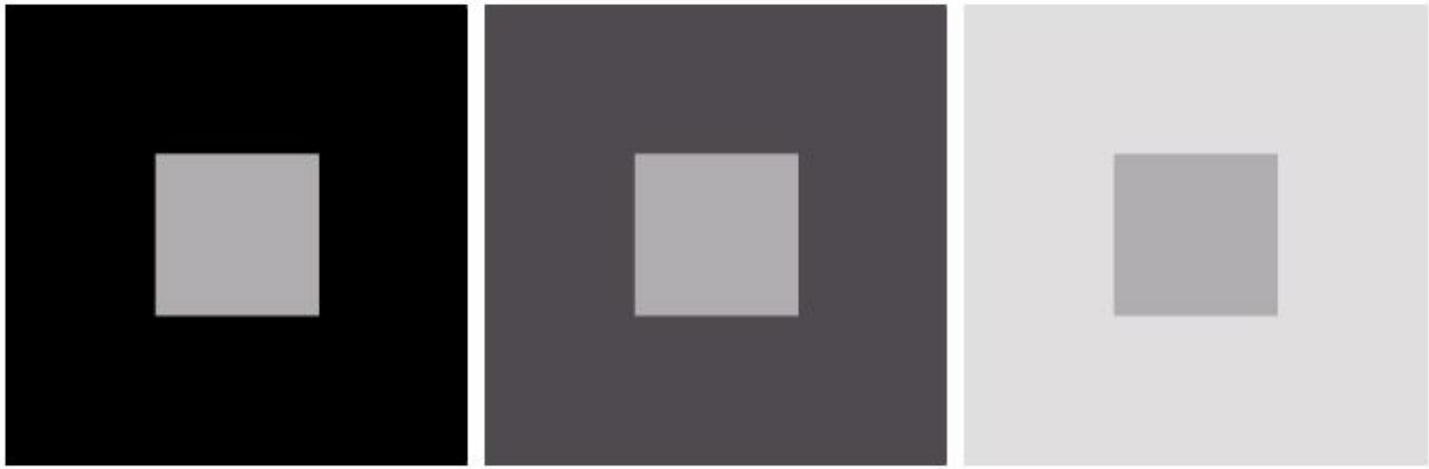
In area A, brightness perceived is darker while in area B is brighter. This phenomenon is called ***Mach Band Effect***.

Brightness Adaptation of Human Eye : Simultaneous Contrast



Simultaneous contrast. All small squares have exactly the same intensity
but they appear progressively darker as background becomes lighter.

Simultaneous Contrast

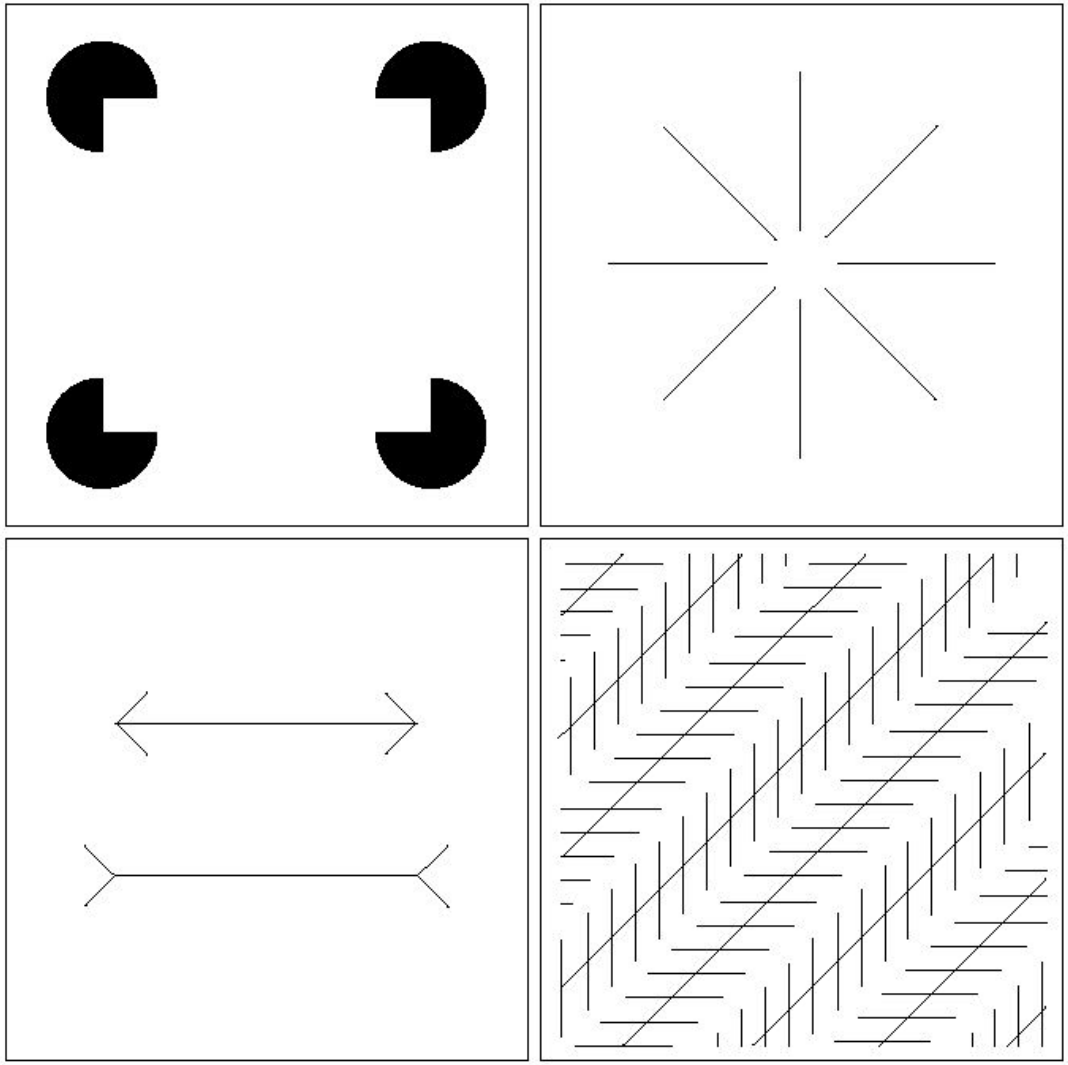


a b c

FIGURE 2.8 Examples of simultaneous contrast. All the inner squares have the same intensity, but they appear progressively darker as the background becomes lighter.

(Images from Rafael C. Gonzalez and Richard E. Wood, Digital Image Processing, 2nd Edition.

Optical illusion



(Images from Rafael C. Gonzalez and Richard E. Wood, Digital Image Processing, 2nd Edition.

Visible Spectrum

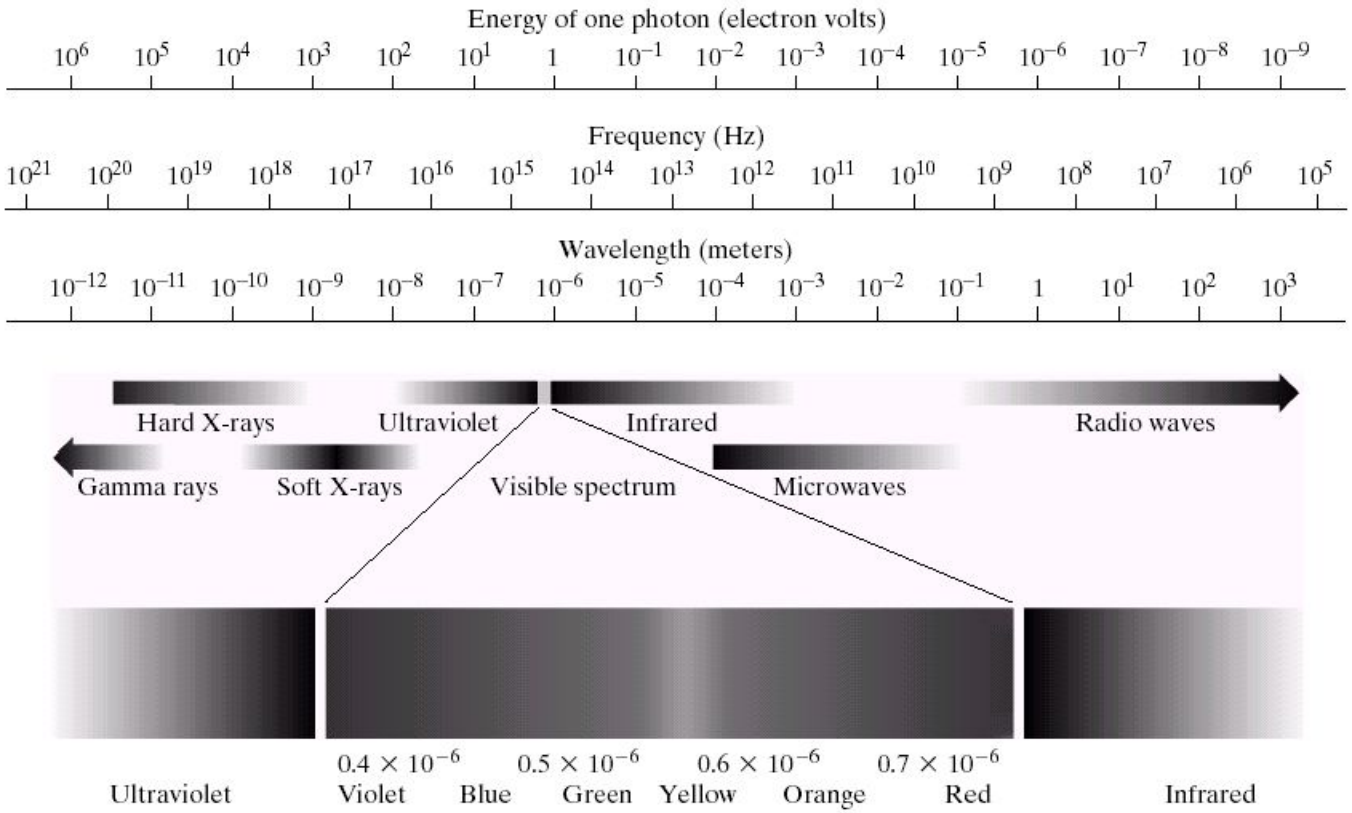
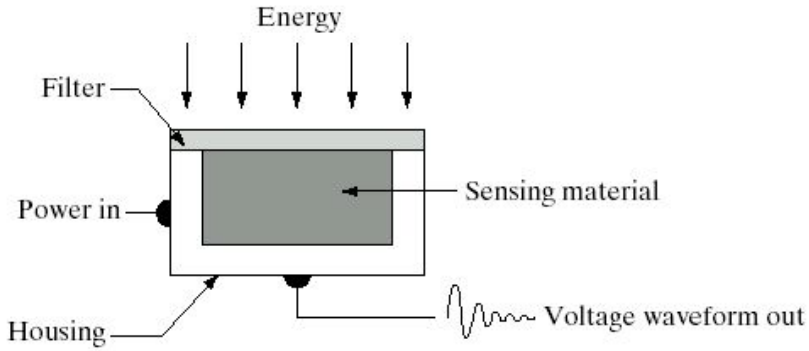


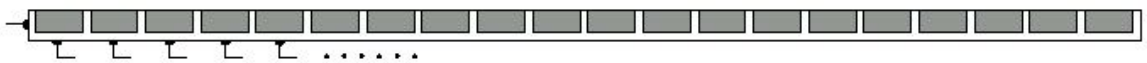
FIGURE 2.10 The electromagnetic spectrum. The visible spectrum is shown zoomed to facilitate explanation, but note that the visible spectrum is a rather narrow portion of the EM spectrum.

(Images from Rafael C. Gonzalez and Richard E. Wood, Digital Image Processing, 2nd Edition.

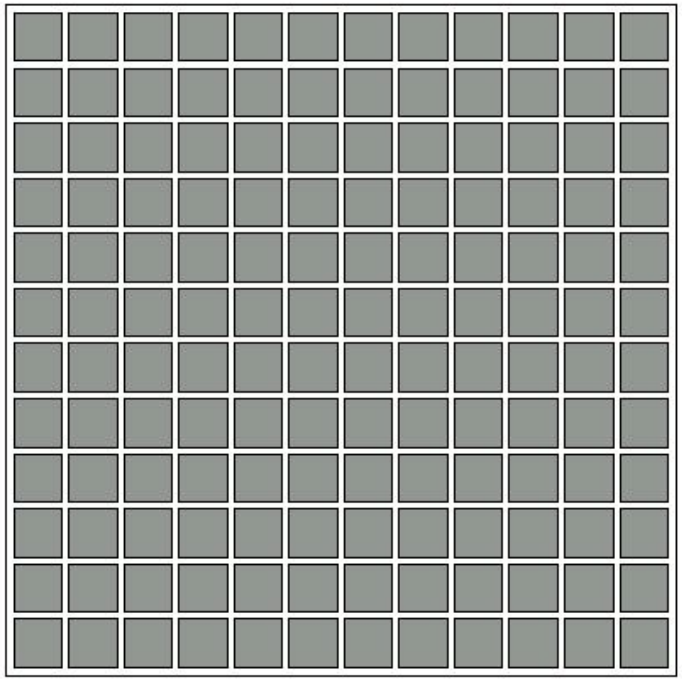
Image Sensors



Single sensor



Line sensor



Array sensor

(Images from Rafael C. Gonzalez and Richard E. Wood, Digital Image Processing, 2nd Edition.

Image Sensors : Single Sensor

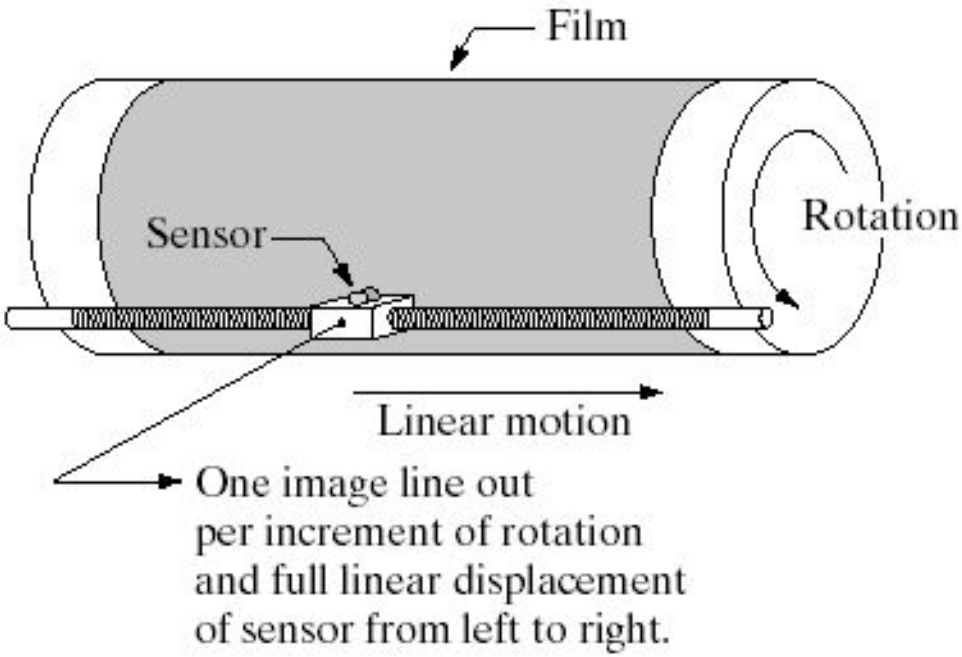
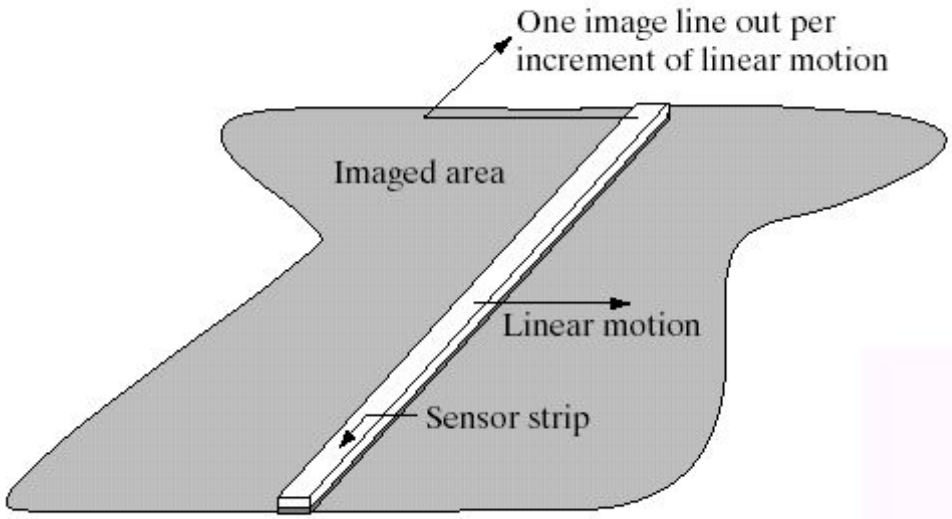


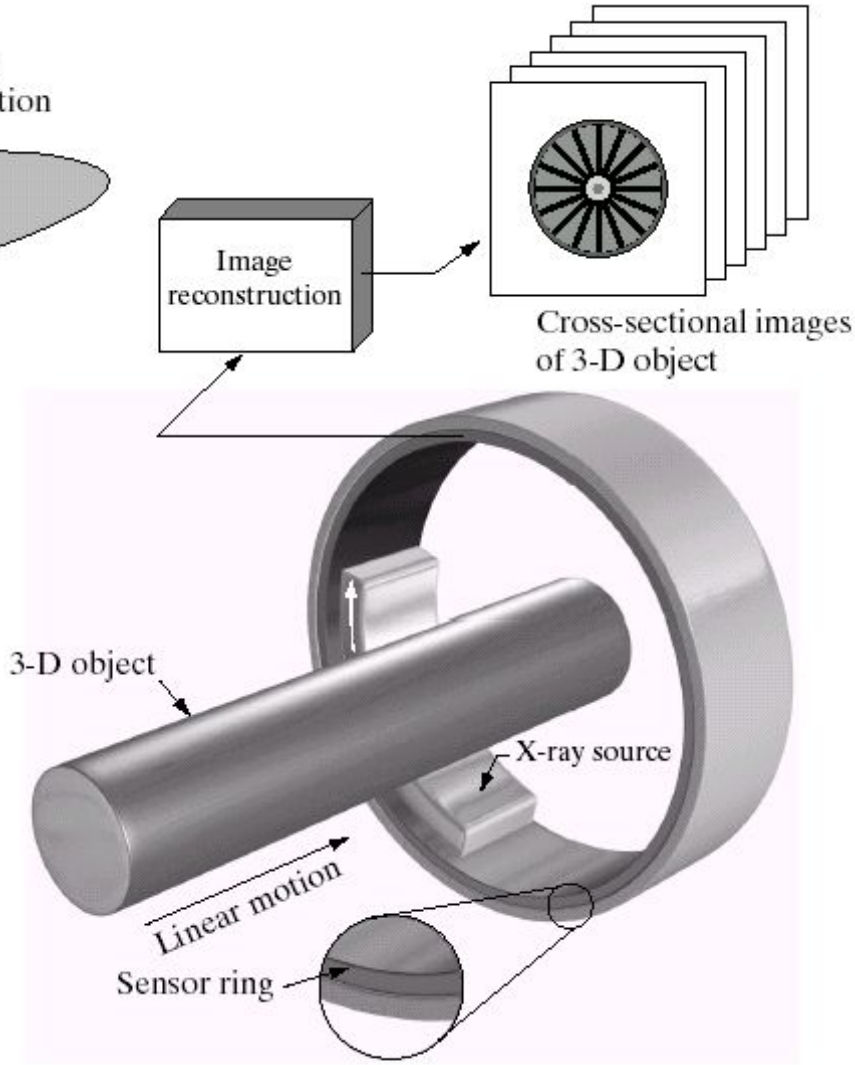
FIGURE 2.13 Combining a single sensor with motion to generate a 2-D image.

(Images from Rafael C. Gonzalez and Richard E. Wood, Digital Image Processing, 2nd Edition.

Image Sensors : Line Sensor



Fingerprint sweep sensor

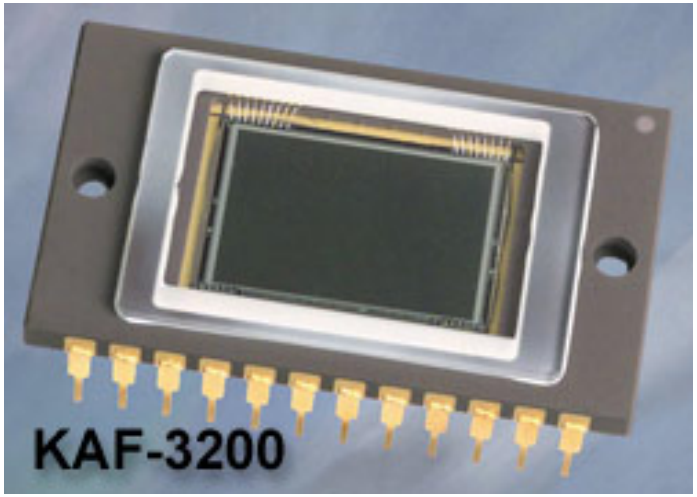


Computerized Axial Tomography

(Images from Rafael C. Gonzalez and Richard E. Wood, Digital Image Processing, 2nd Edition.

Image Sensors : Array Sensor

Charge-Coupled Device (CCD)



CCD KAF-3200E from Kodak.
(2184 x 1472 pixels,
Pixel size 6.8 microns^2)

- Used for convert a continuous image into a digital image
- Contains an array of light sensors
- Converts photon into electric charges accumulated in each sensor unit

Image Sensor: Inside Charge-Coupled Device

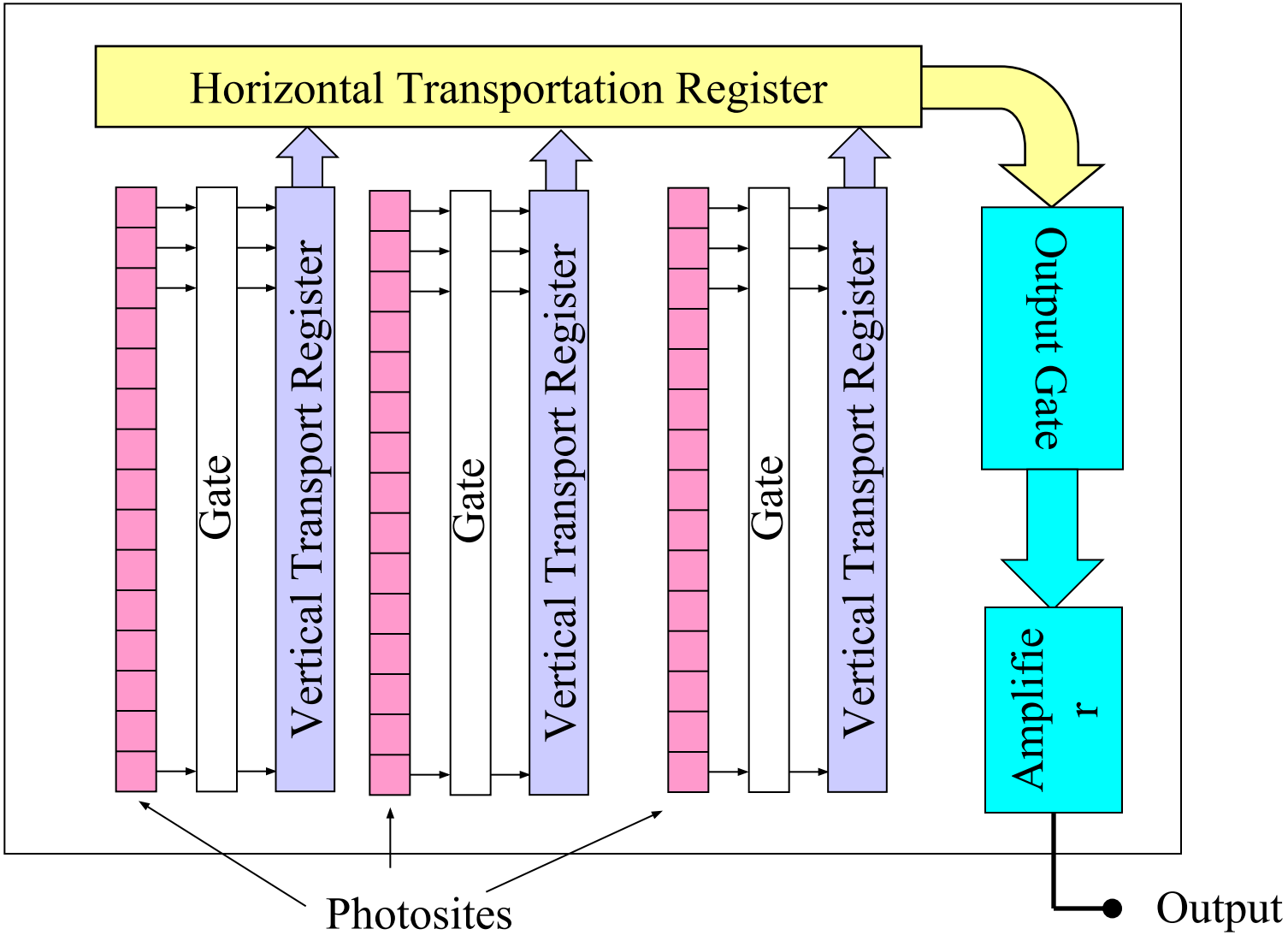
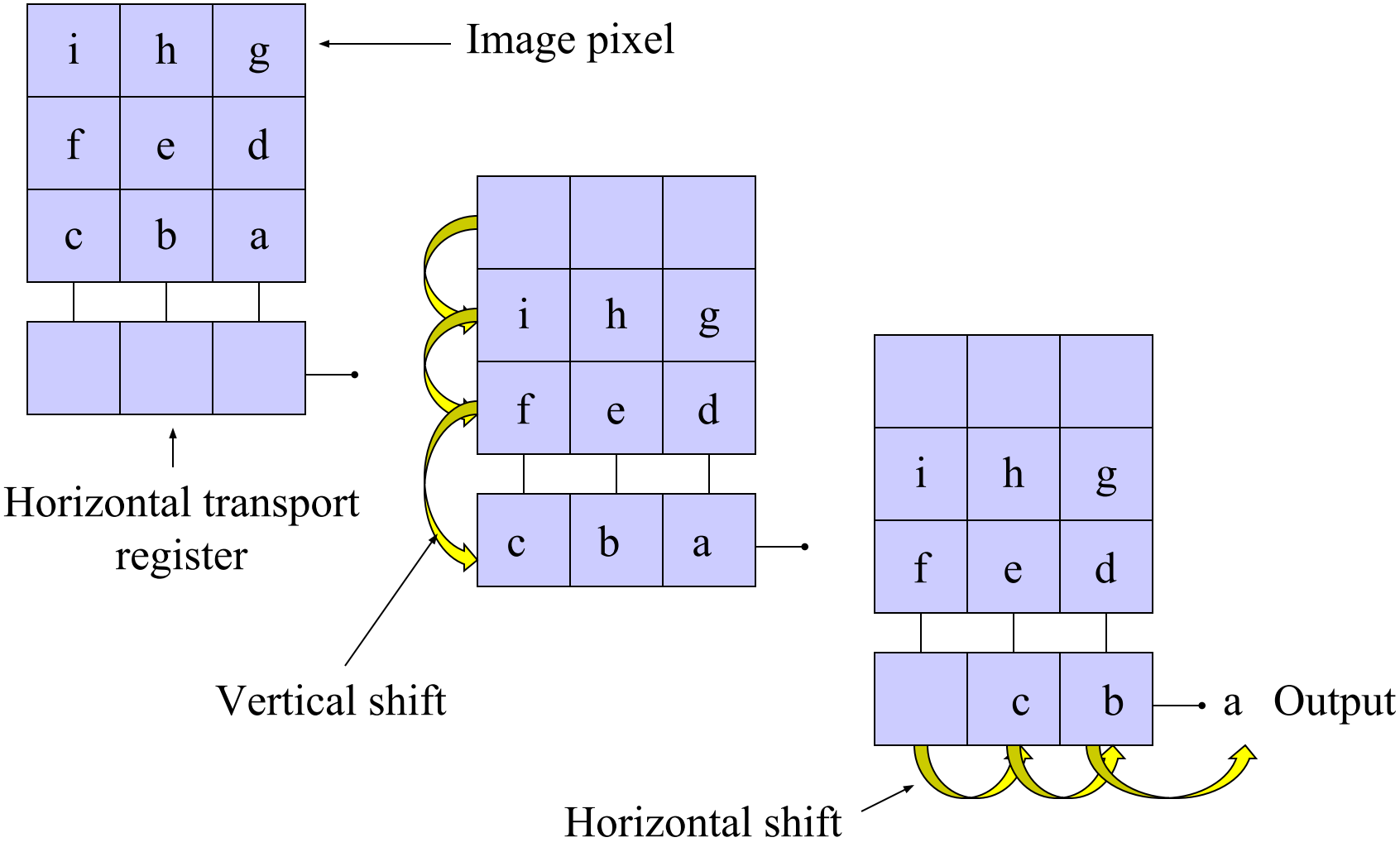


Image Sensor: How CCD works



Fundamentals of Digital Images

- **An image**: **a multidimensional function of spatial coordinates**.
- **Spatial coordinate**: (x,y) for 2D case such as photograph,
  (x,y,z) for 3D case such as CT scan images
  (x,y,t) for movies
- The function f may represent intensity (for monochrome images) or color (for color images) or other associated values.

Digital Images

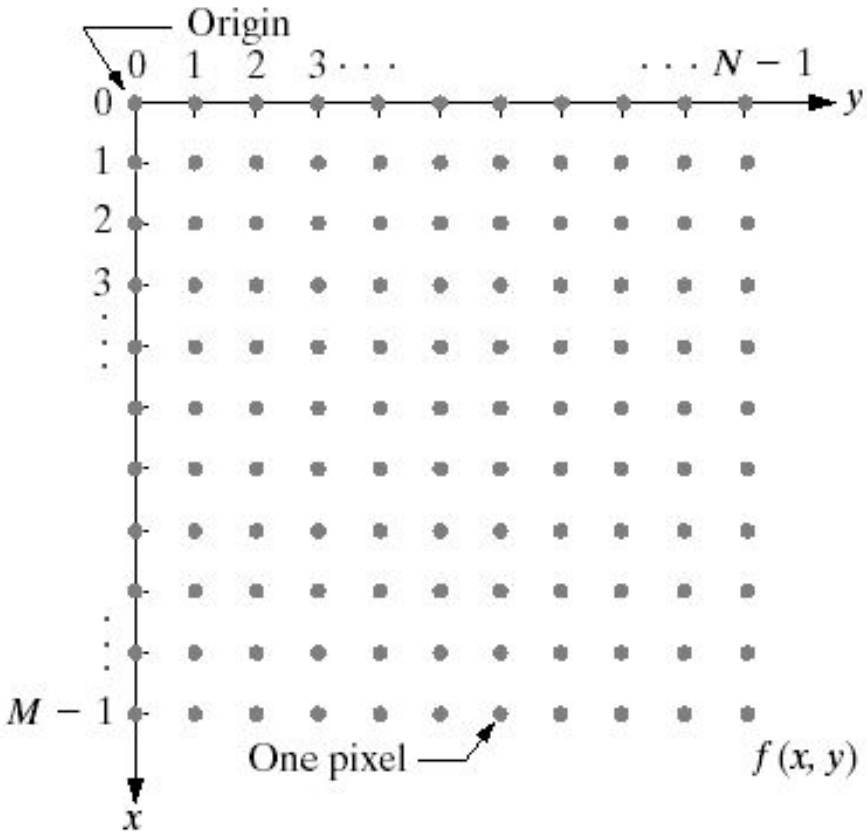
Digital image: an image that has been discretized both in Spatial coordinates and associated value.

- Consist of 2 sets:(1) a point set and (2) a value set

- Can be represented in the form
  $I = \{(x,a(x)): x \in X, a(x) \in F\}$
  where X and F are a point set and value set, respectively.

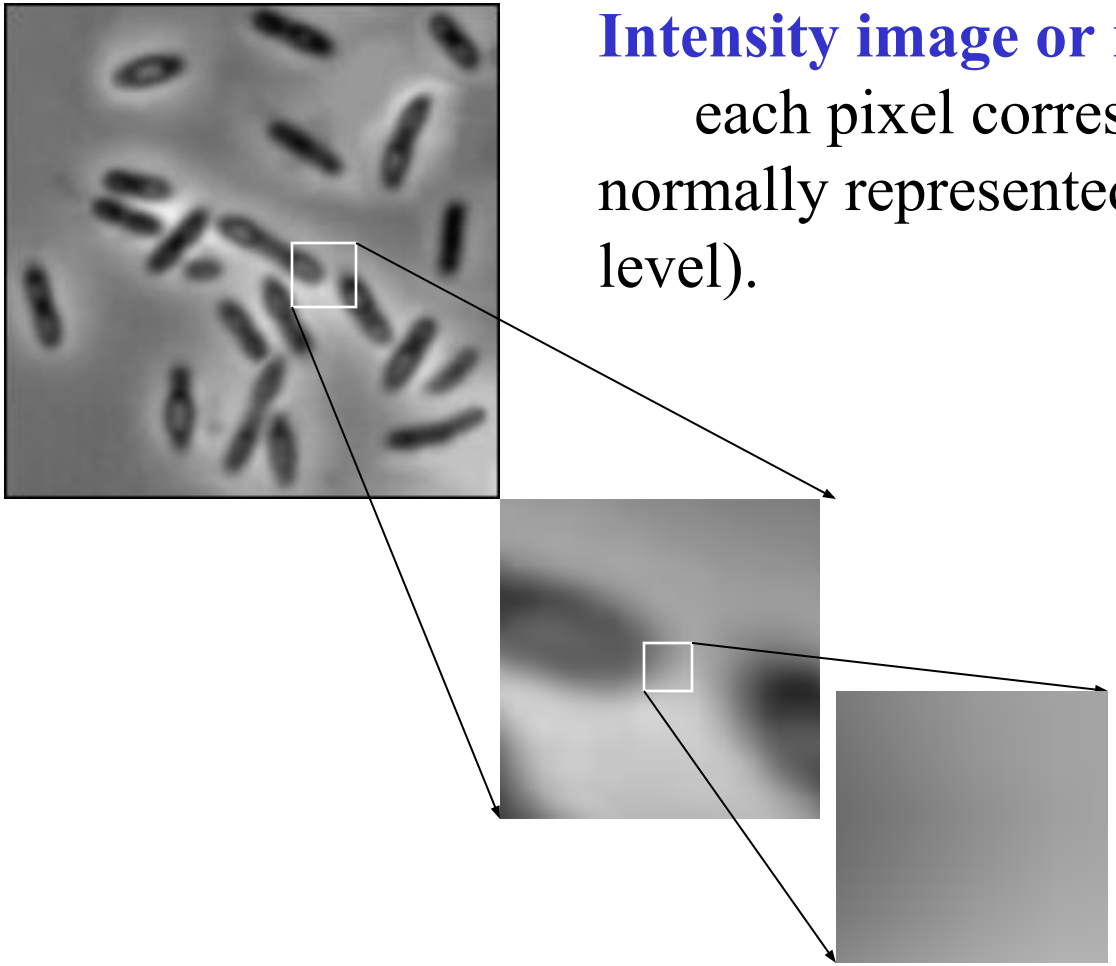
- An element of the image, $(x,a(x))$ is called a ***pixel*** where
  - x is called the pixel location and
  - $a(x)$ is the pixel value at the location x

Conventional Coordinate for Image Representation



(Images from Rafael C. Gonzalez and Richard E. Wood, Digital Image Processing, 2nd Edition.

Digital Image Types : Intensity Image



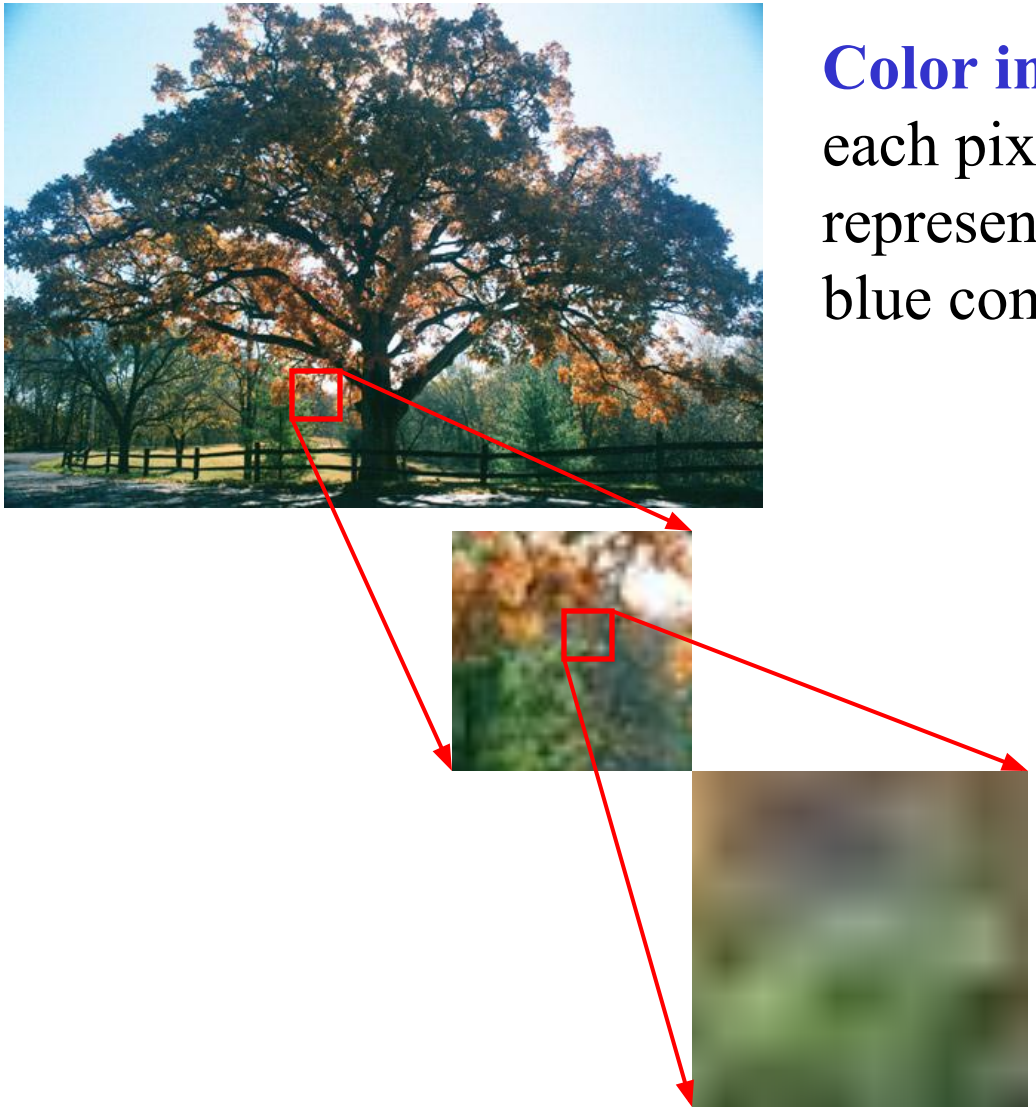
Intensity image or monochrome image

each pixel corresponds to light intensity normally represented in gray scale (gray level).

Gray scale values

$$\begin{bmatrix} 10 & 10 & 16 & 28 \\ 9 & 6 & 26 & 37 \\ 15 & 25 & 13 & 22 \\ 32 & 15 & 87 & 39 \end{bmatrix}$$

Digital Image Types : RGB Image



Color image or RGB image: each pixel contains a vector representing red, green and blue components.

RGB components

$$\begin{bmatrix} 10 & 10 & 16 & 28 \\ 9 & \begin{bmatrix} 65 & 70 & 56 & 43 \\ 32 & 25 & 60 & 54 \\ 21 & 15 & 85 & 65 \\ 54 & & 32 & 65 \end{bmatrix} & \begin{bmatrix} 6 & 26 & 37 & 56 \\ 99 & 54 & 96 & 67 \\ 13 & 90 & 22 & 96 \\ 87 & 47 & 39 & 43 \\ 85 & 65 & 87 & 99 \end{bmatrix} & \begin{bmatrix} 78 \\ 67 \\ 42 \\ 92 \\ 39 \end{bmatrix} \\ 15 & & & \\ 32 & & & \end{bmatrix}$$

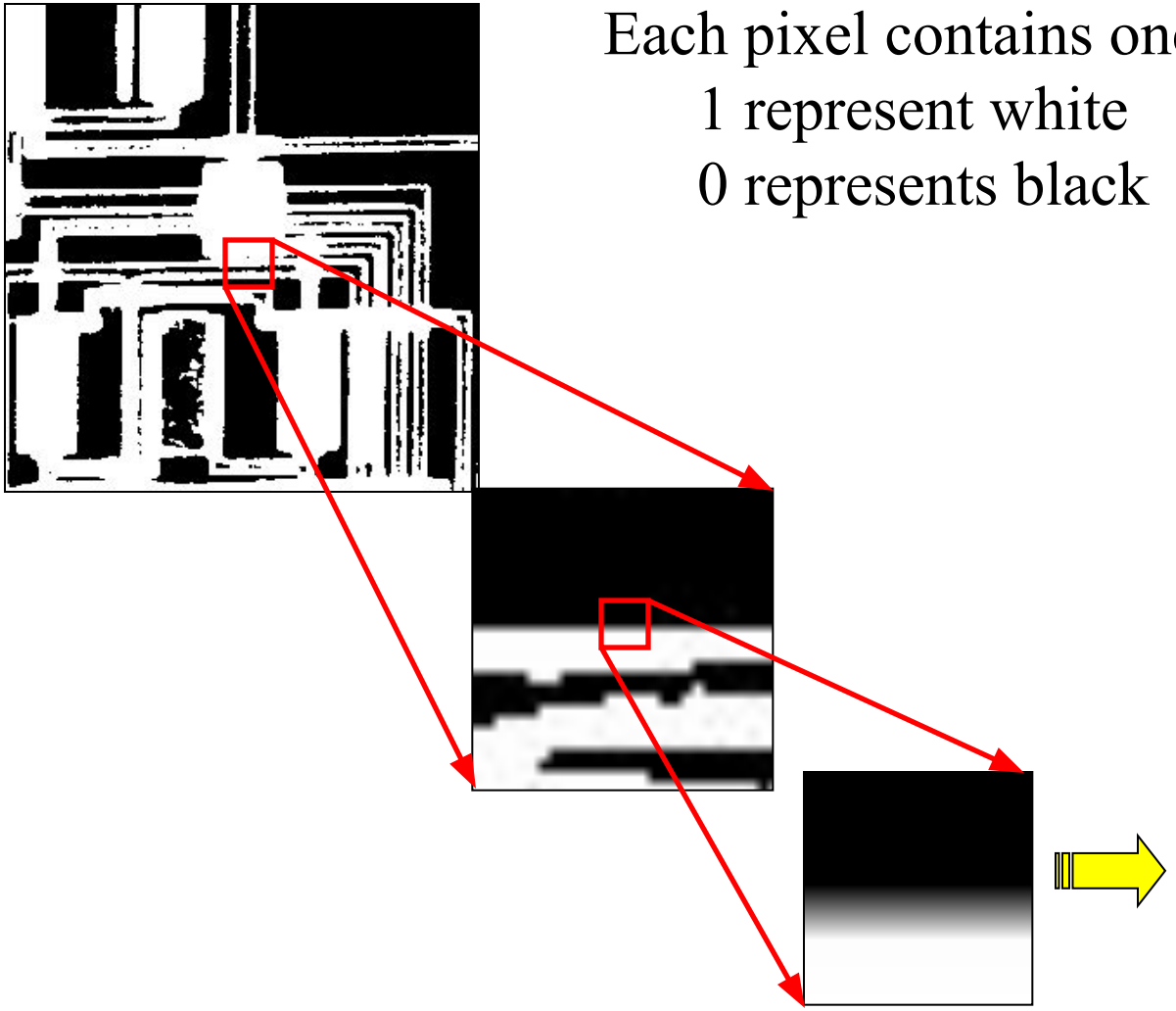
Image Types : Binary Image

Binary image or black and white image

Each pixel contains one bit :

1 represent white

0 represents black



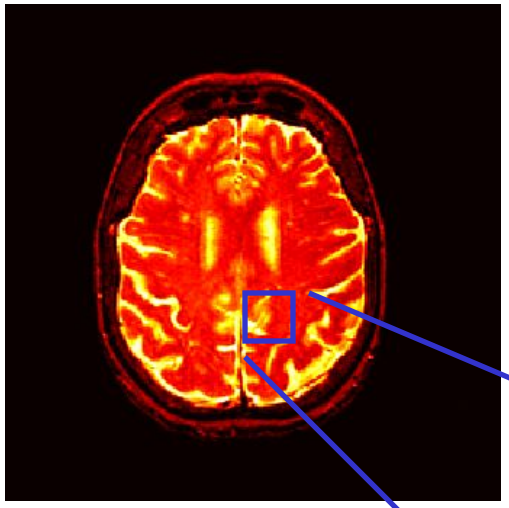
Binary data

$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

Image Types : Index Image

Index image

Each pixel contains index number pointing to a color in a color table



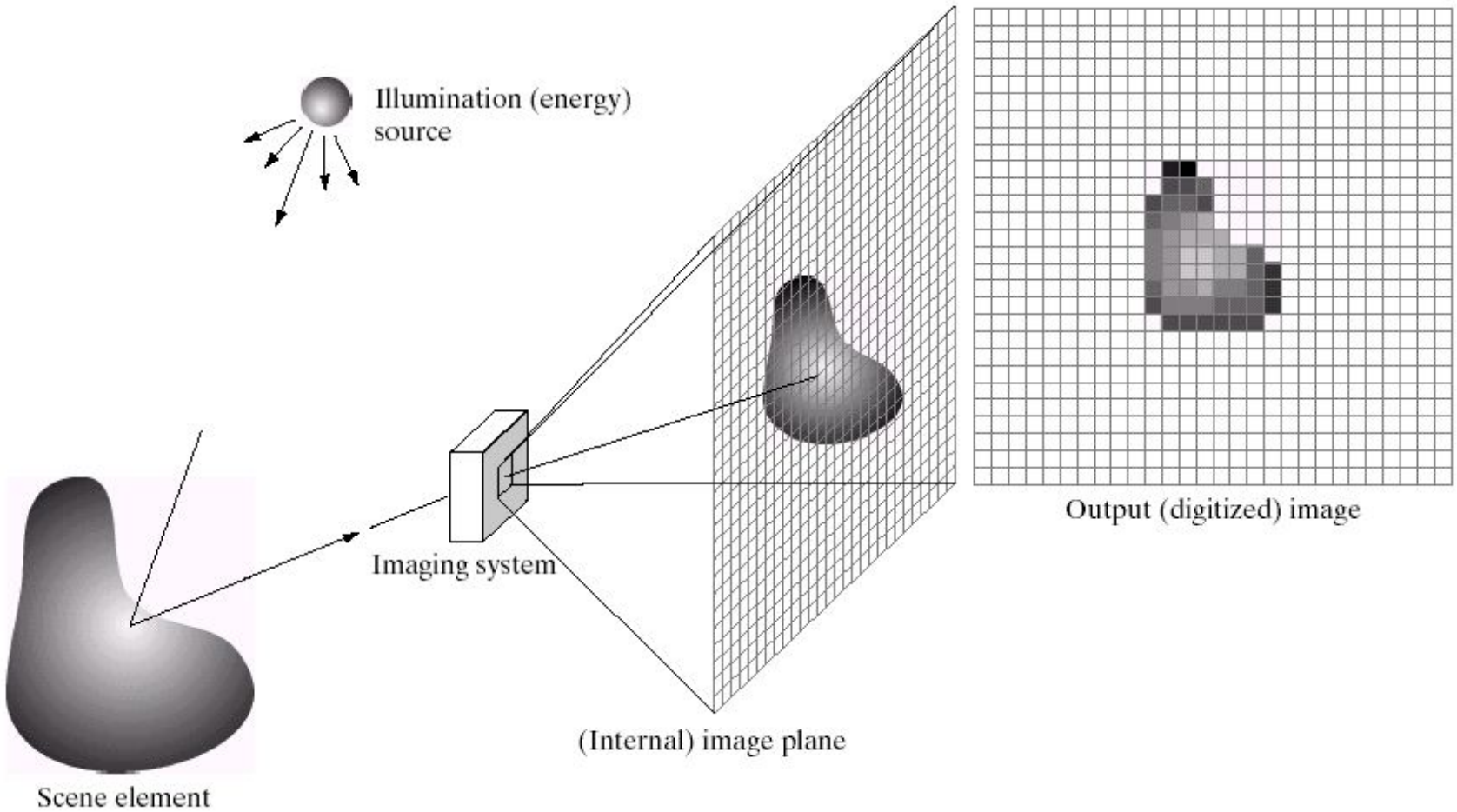
$$\begin{bmatrix} 1 & 4 & 9 \\ 6 & 4 & 7 \\ 6 & 5 & 2 \end{bmatrix}$$

Index value

Color Table

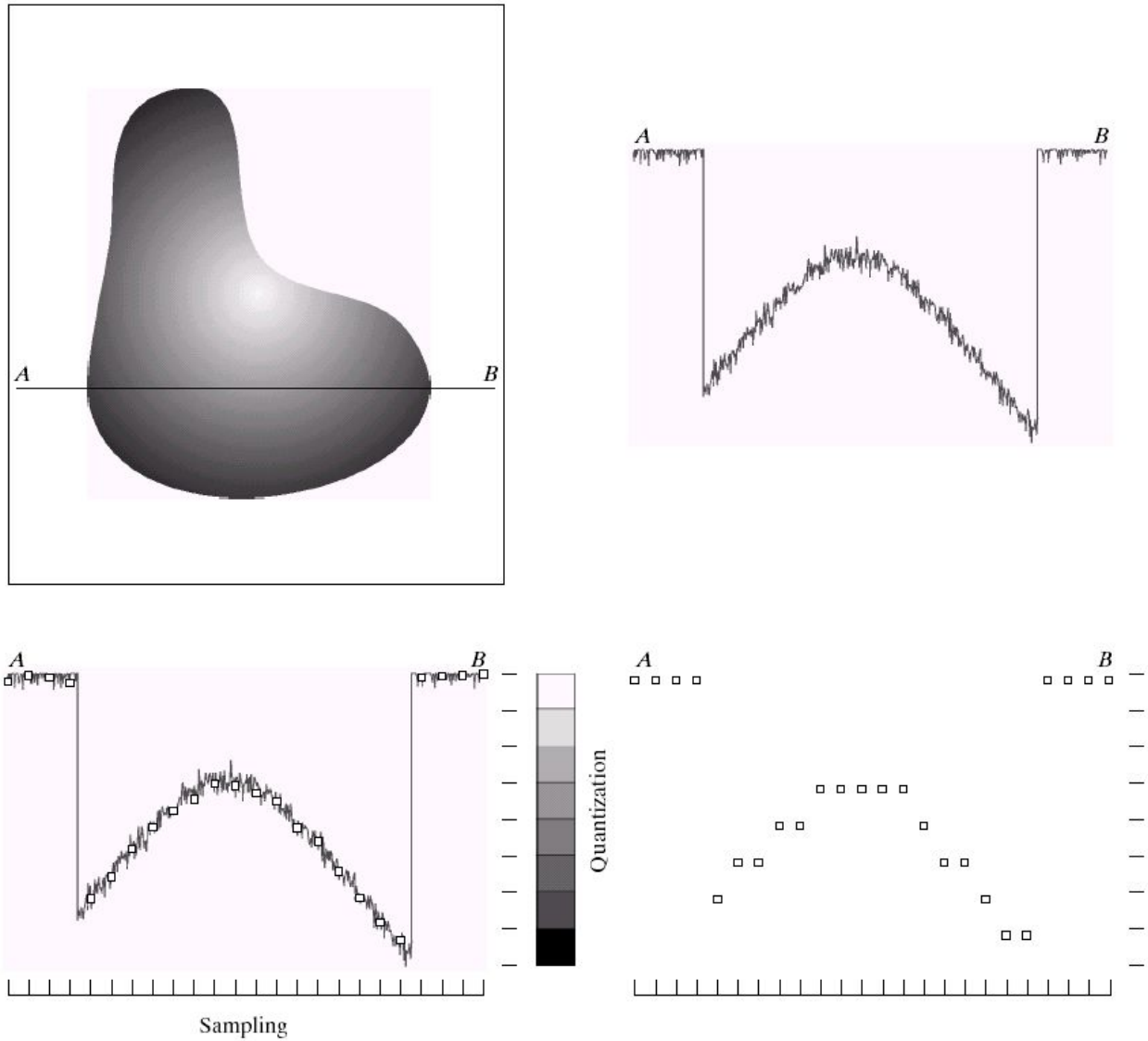
Index No.	Red component	Green component	Blue component
1	0.1	0.5	0.3
2	1.0	0.0	0.0
3	0.0	1.0	0.0
4	0.5	0.5	0.5
5	0.2	0.8	0.9
…	…	…	…

Digital Image Acquisition Process



(Images from Rafael C. Gonzalez and Richard E. Wood, Digital Image Processing, 2nd Edition.

Generating a Digital Image

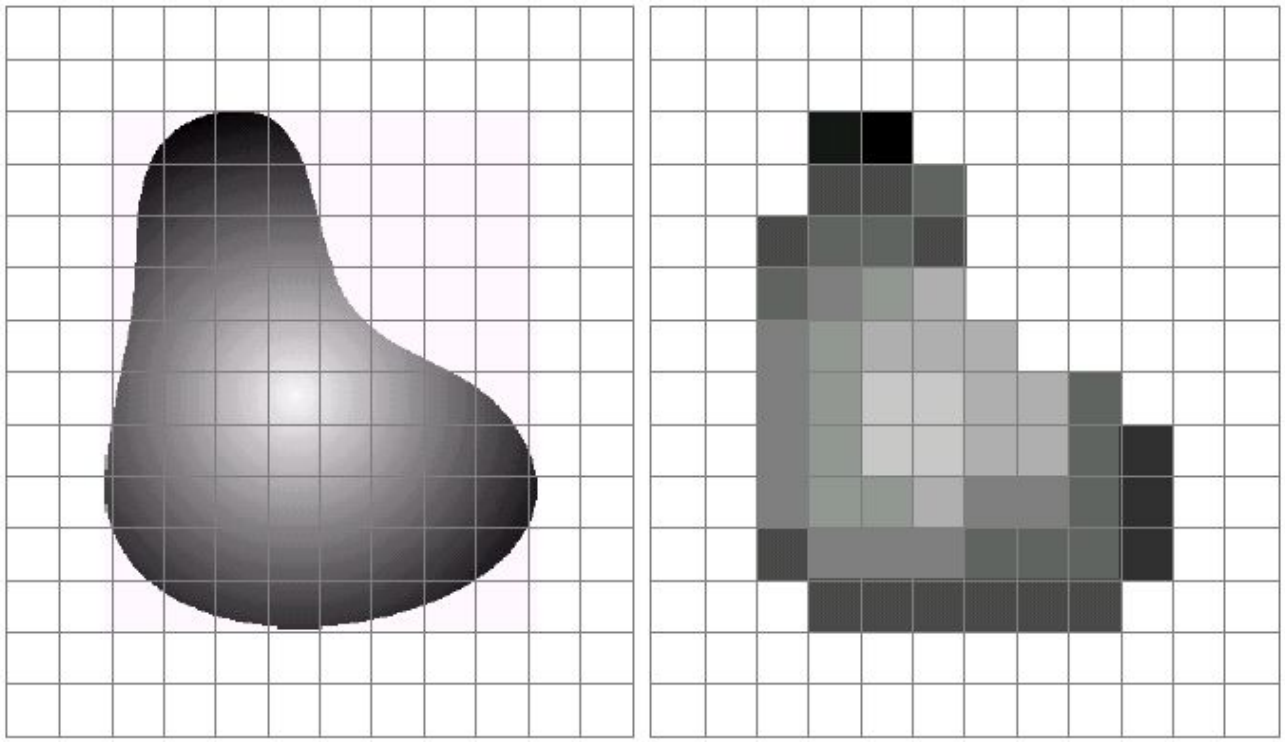


(Images from Rafael C. Gonzalez and Richard E. Wood, Digital Image Processing, 2nd Edition.

a b
c d

FIGURE 2.16 Generating a digital image. (a) Continuous image. (b) A scan line from *A* to *B* in the continuous image, used to illustrate the concepts of sampling and quantization. (c) Sampling and quantization. (d) Digital scan line.

Image Sampling and Quantization



a b

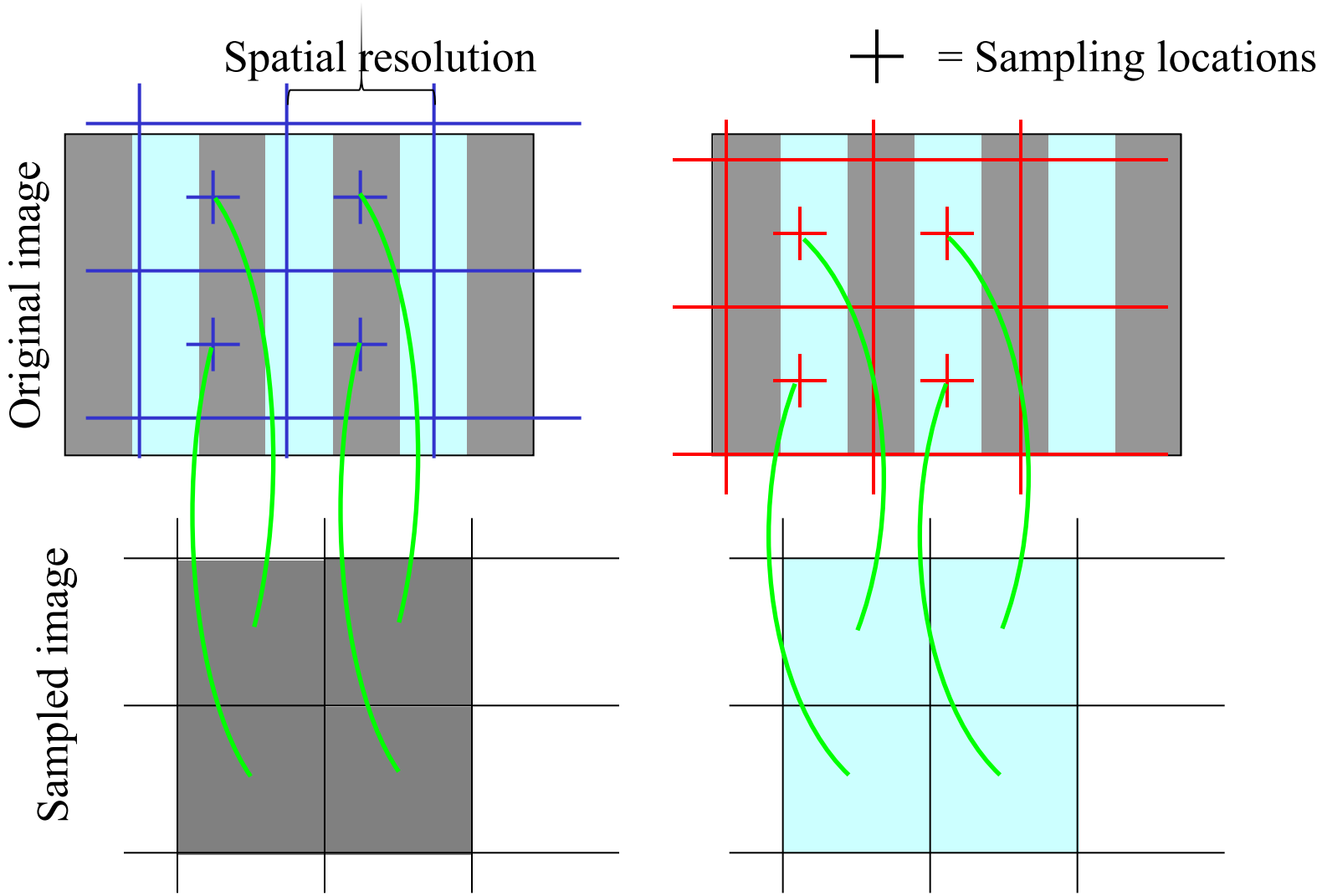
FIGURE 2.17 (a) Continuos image projected onto a sensor array. (b) Result of image sampling and quantization.

Image sampling: discretize an image in the spatial domain

Spatial resolution / image resolution: pixel size or number of pixels

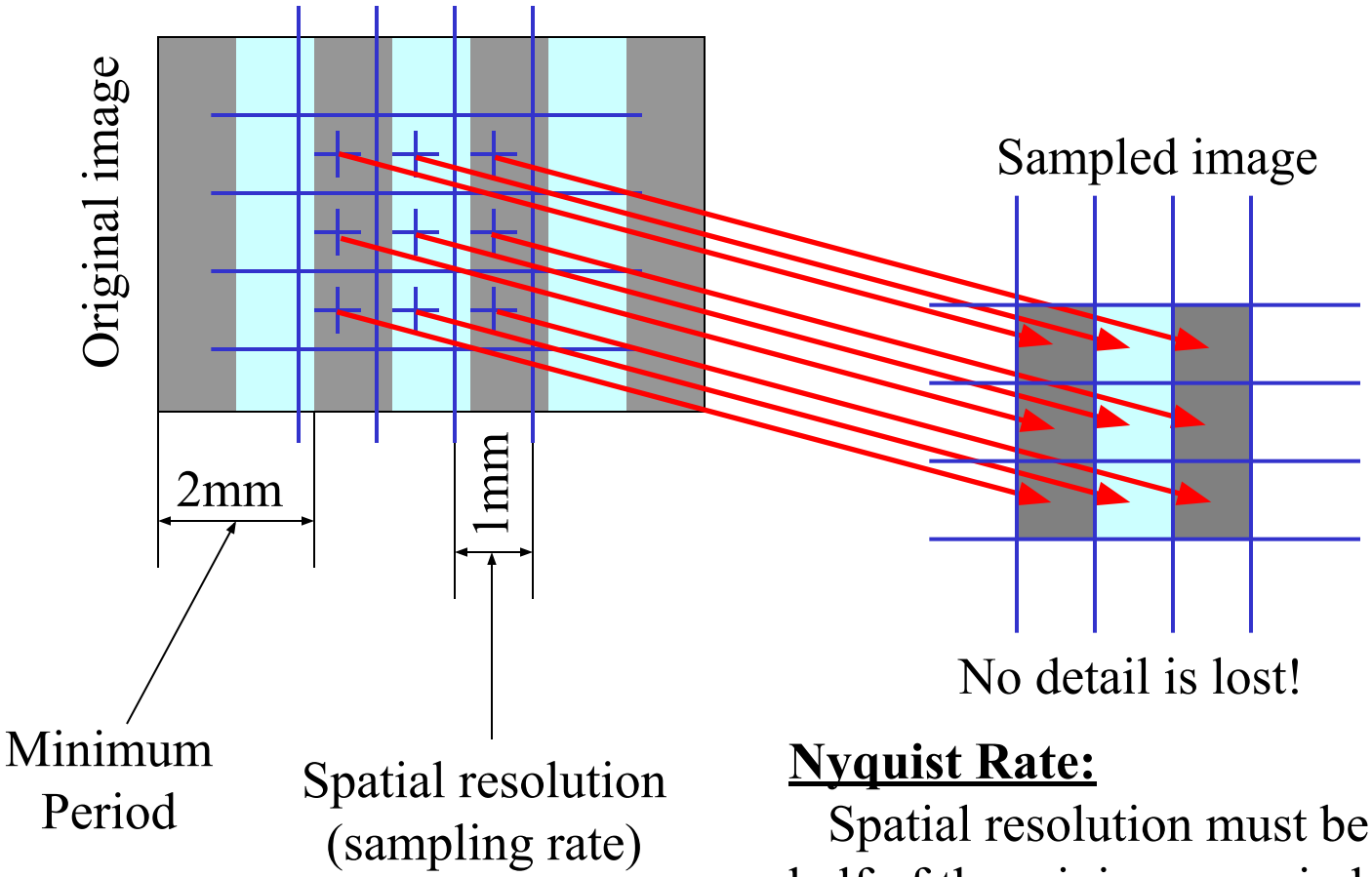
(Images from Rafael C. Gonzalez and Richard E. Wood, Digital Image Processing, 2nd Edition.

How to choose the spatial resolution



Under sampling, we lost some image details!

How to choose the spatial resolution : Nyquist Rate



Nyquist Rate:

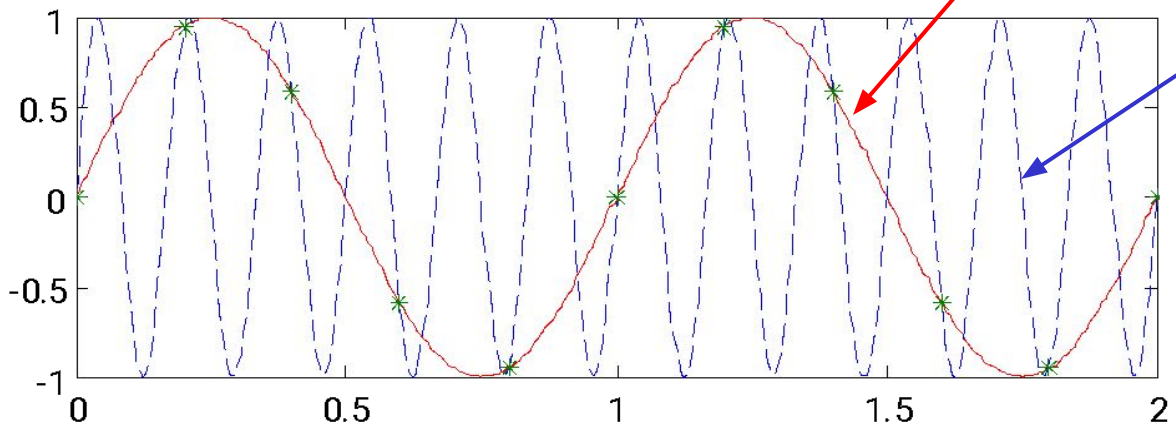
Spatial resolution must be less or equal half of the minimum period of the image or sampling frequency must be greater or Equal twice of the maximum frequency.

+ = Sampling locations

Aliased Frequency

$$x_1(t) = \sin(2\pi t), \qquad f = 1$$

$$x_2(t) = \sin(12\pi t), \qquad f = 6$$

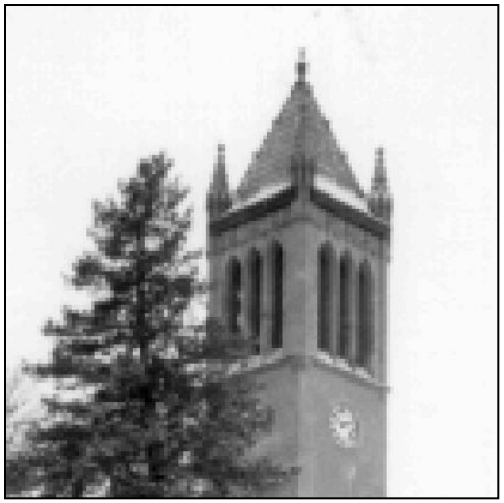


Sampling rate:
5 samples/sec

Two different frequencies but the same results !

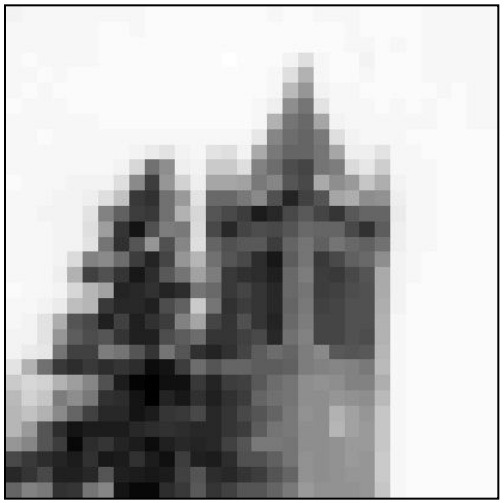
Effect of Spatial Resolution

256x256 pixels



128x128 pixels

64x64 pixels



32x32 pixels

Effect of Spatial Resolution

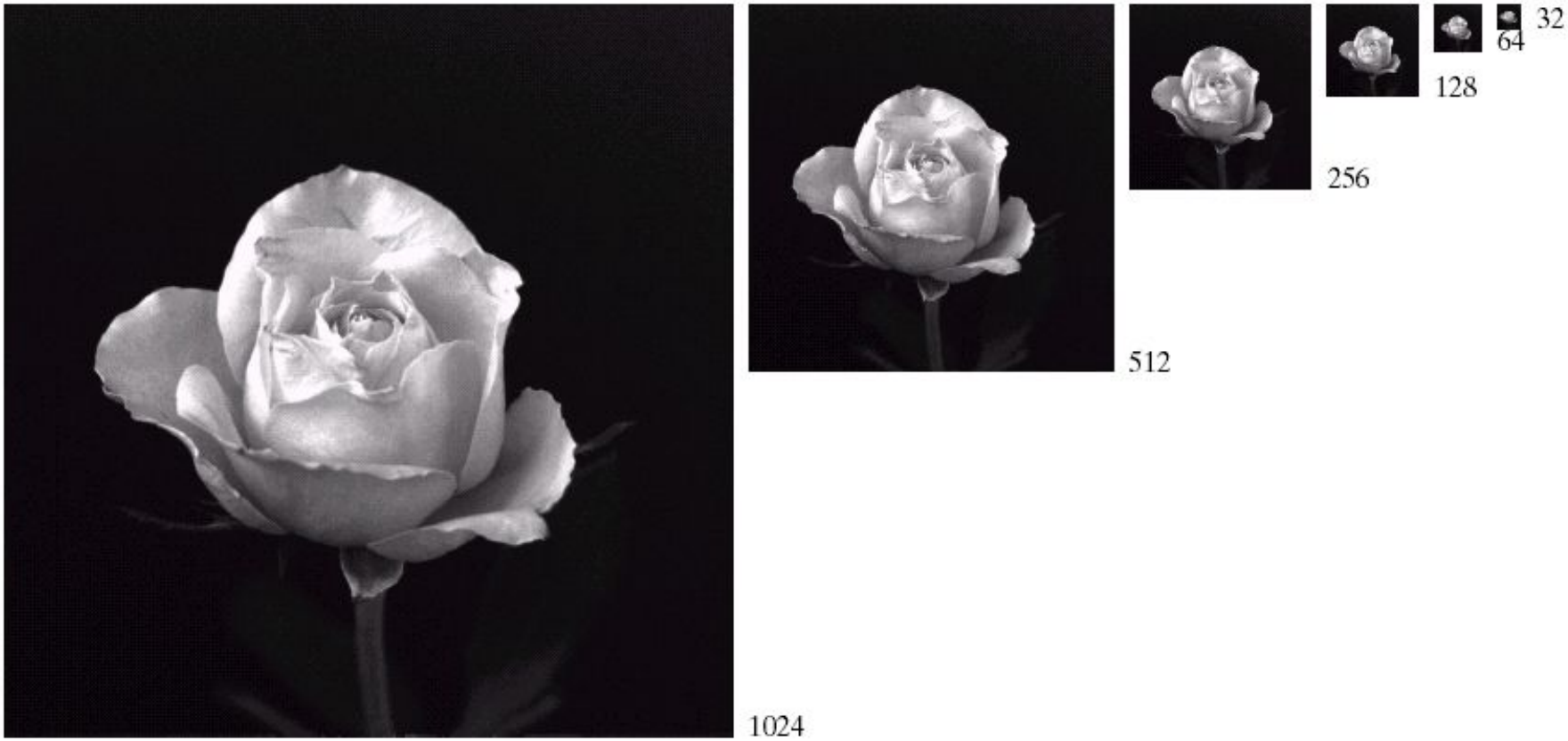
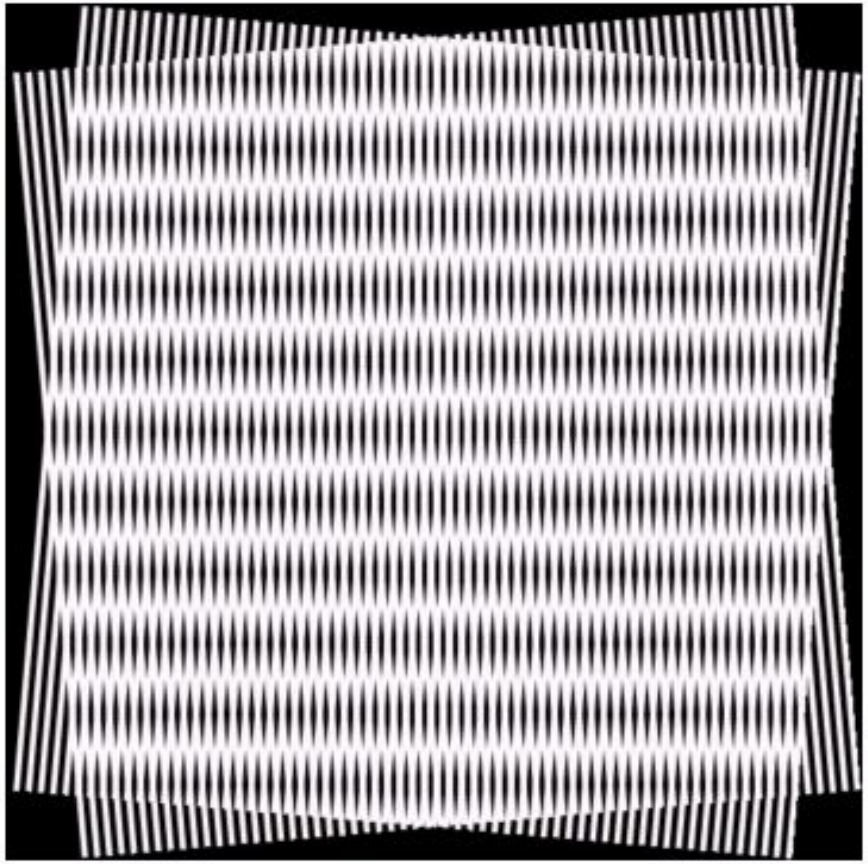


FIGURE 2.19 A 1024 × 1024, 8-bit image subsampled down to size 32 × 32 pixels. The number of allowable gray levels was kept at 256.

(Images from Rafael C. Gonzalez and Richard E. Wood, Digital Image Processing, 2nd Edition.

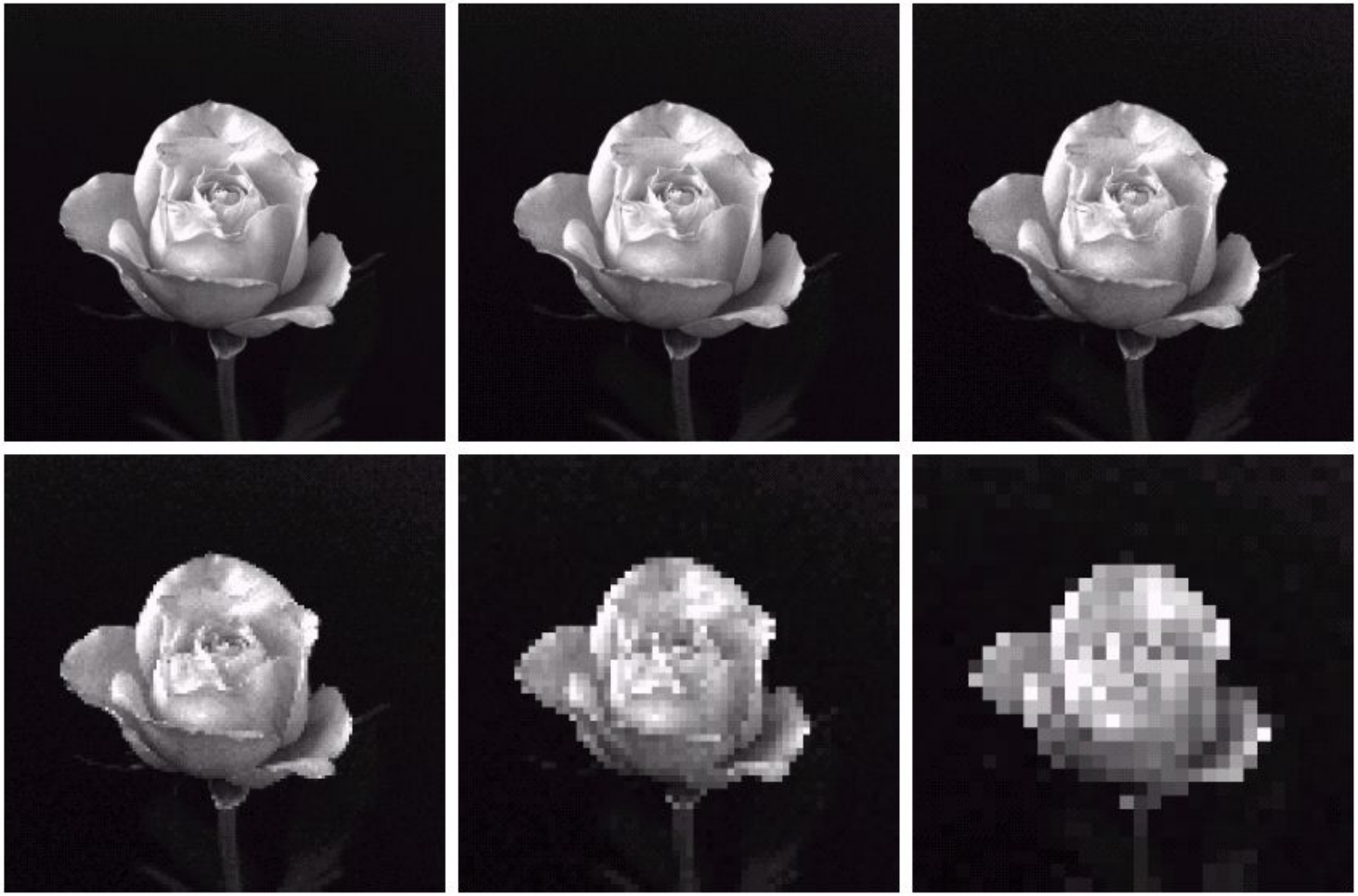
Moire Pattern Effect : Special Case of Sampling



Moire patterns occur when frequencies of two superimposed periodic patterns are close to each other.

(Images from Rafael C. Gonzalez and Richard E. Wood, Digital Image Processing, 2nd Edition.

Effect of Spatial Resolution



a b c
d e f

FIGURE 2.20 (a) 1024 × 1024, 8-bit image. (b) 512 × 512 image resampled into 1024 × 1024 pixels by row and column duplication. (c) through (f) 256 × 256, 128 × 128, 64 × 64, and 32 × 32 images resampled into 1024 × 1024 pixels.

(Images from Rafael C. Gonzalez and Richard E. Wood, Digital Image Processing, 2nd Edition.

Can we increase spatial resolution by interpolation ?

a b c
d e f

FIGURE 2.25 Top row: images zoomed from 128 × 128, 64 × 64, and 32 × 32 pixels to 1024 × 1024 pixels, using nearest neighbor gray-level interpolation. Bottom row: same sequence, but using bilinear interpolation.

Down sampling is an irreversible process.

(Images from Rafael C. Gonzalez and Richard E. Wood, Digital Image Processing, 2nd Edition.

Image Quantization

Image quantization:

discretize continuous pixel values into discrete numbers

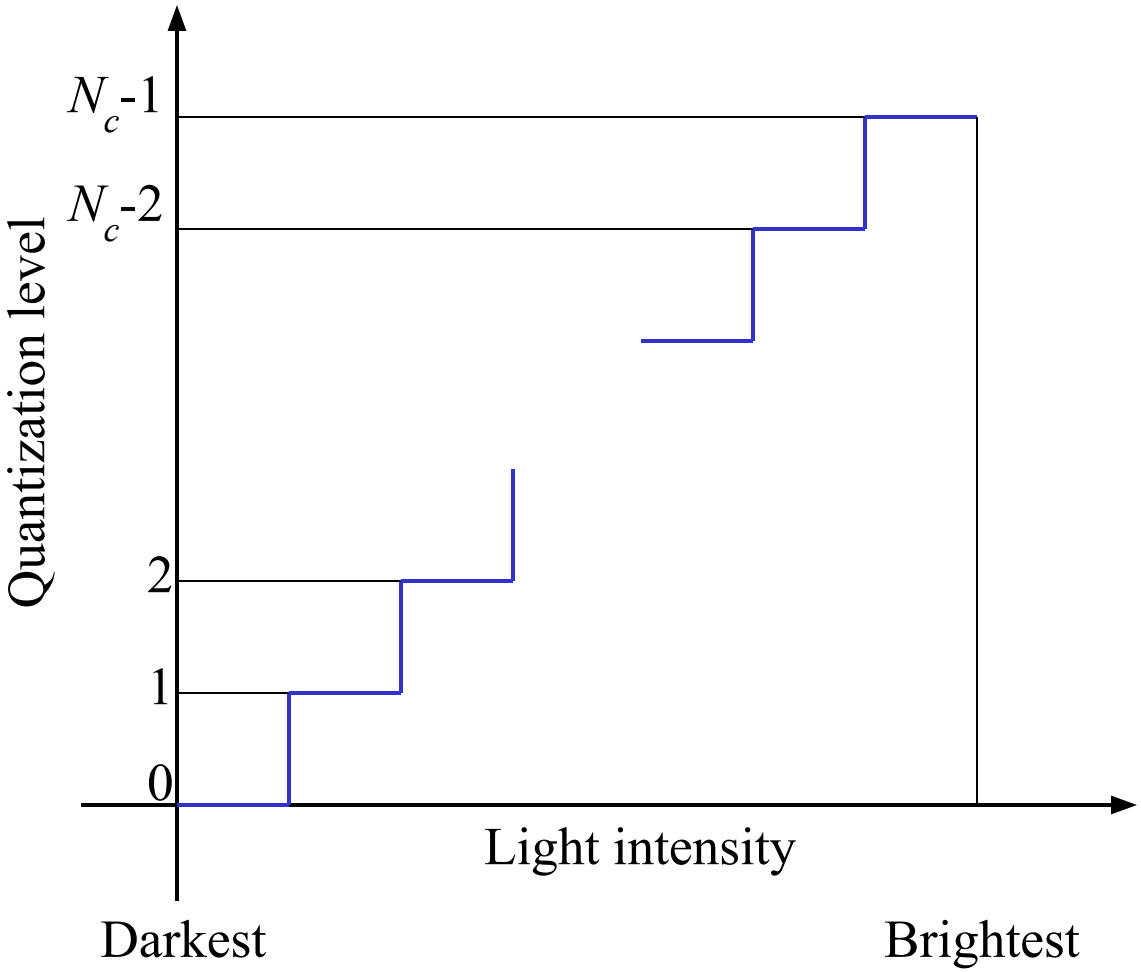
Color resolution/ color depth/ levels:

- No. of colors or gray levels or
- No. of bits representing each pixel value
- No. of colors or gray levels N_c is given by

$$N_c = 2^b$$

where b = no. of bits

Quantization function



Effect of Quantization Levels

256 levels

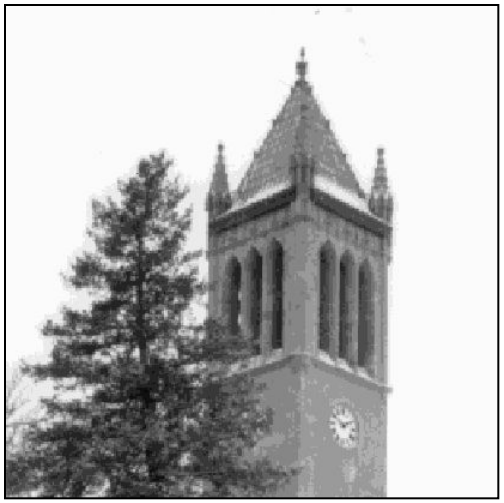
128 levels

64 levels

32 levels

Effect of Quantization Levels (cont.)

16 levels



8 levels

In this image, it is easy to see false contour.

4 levels

2 levels

How to select the suitable size and pixel depth of images

The word “suitable” is subjective: depending on “subject”.

Low detail image Medium detail image High detail image

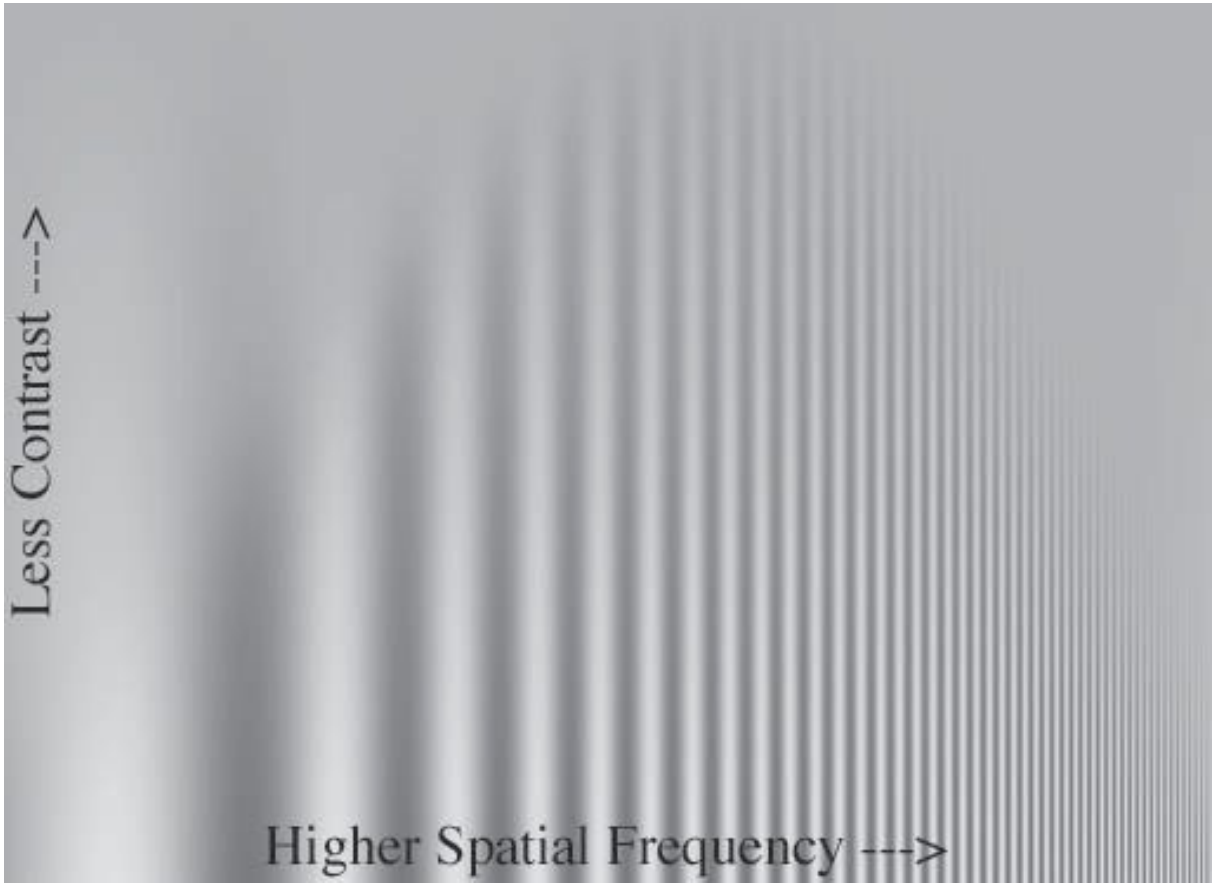
Lena image Cameraman image

To satisfy human mind

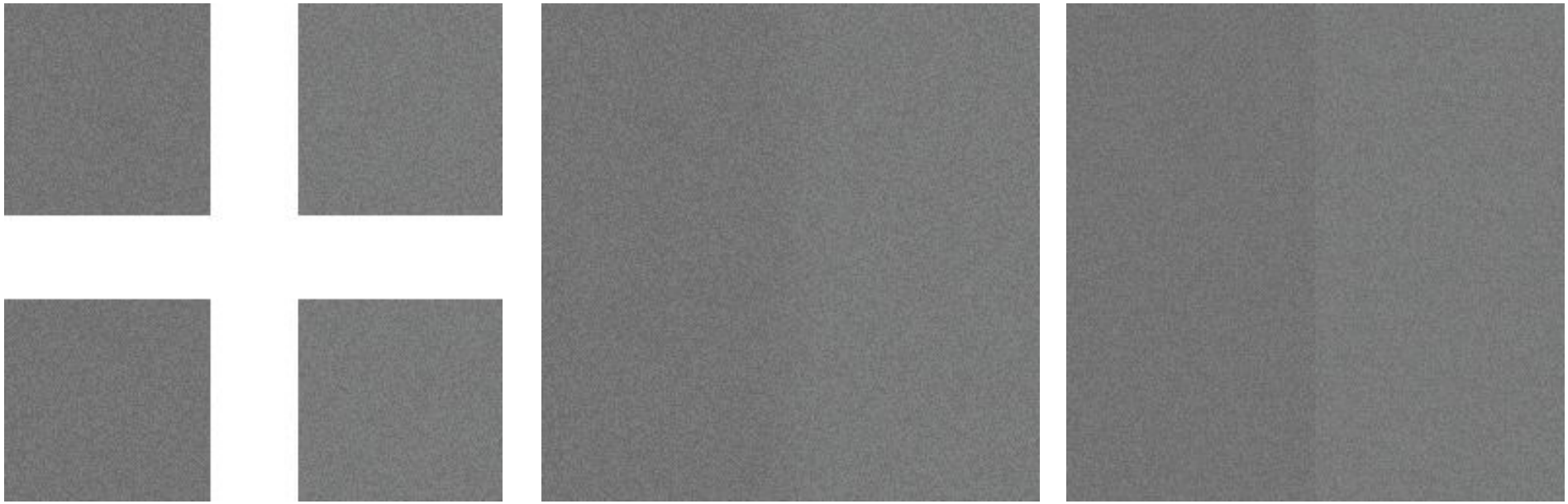
1. For images of the same size, the low detail image may need more pixel depth.
2. As an image size increase, fewer gray levels may be needed.

(Images from Rafael C. Gonzalez and Richard E. Wood, Digital Image Processing, 2nd Edition.

Human vision: Spatial Frequency vs Contrast

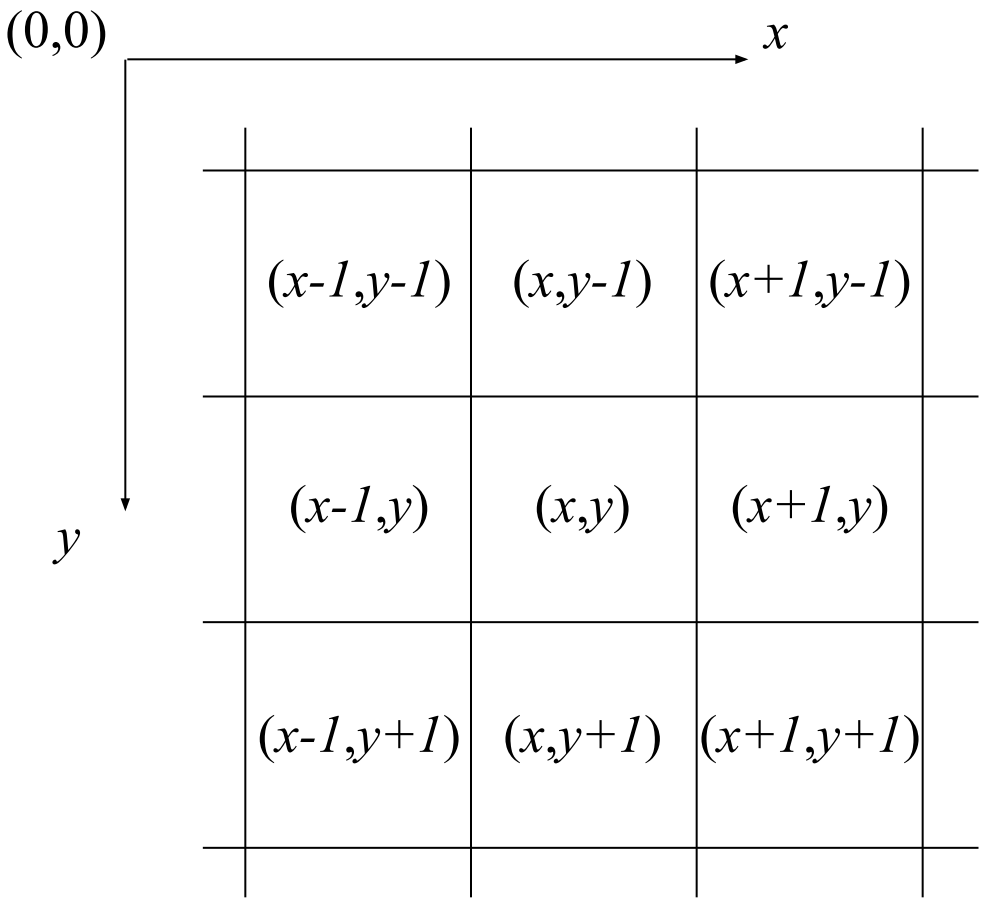


Human vision: Distinguish ability for Difference in brightness



Regions with 5% brightness difference

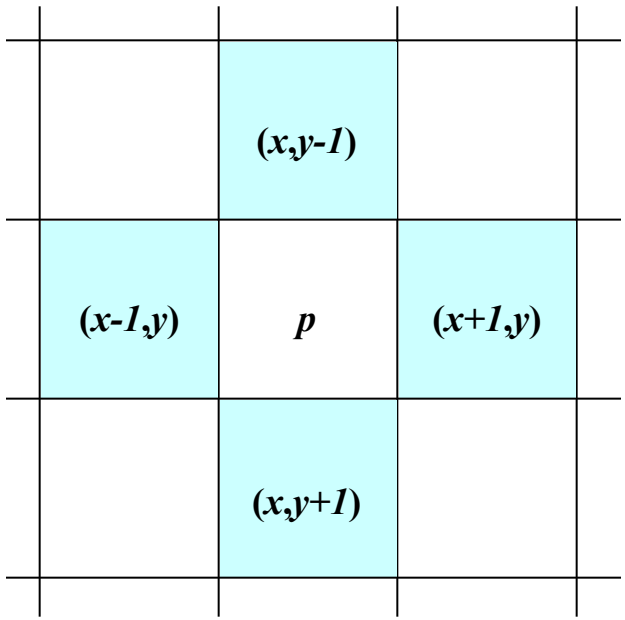
Basic Relationship of Pixels



Conventional indexing method

Neighbors of a Pixel

Neighborhood relation is used to tell adjacent pixels. It is useful for analyzing regions.



4-neighbors of *p*:

$$N_4(p) = \left\{ \begin{array}{l} (x-1,y) \\ (x+1,y) \\ (x,y-1) \\ (x,y+1) \end{array} \right\}$$

4-neighborhood relation considers only vertical and horizontal neighbors.

Note: $q \in N_4(p)$ implies $p \in N_4(q)$

Neighbors of a Pixel (cont.)

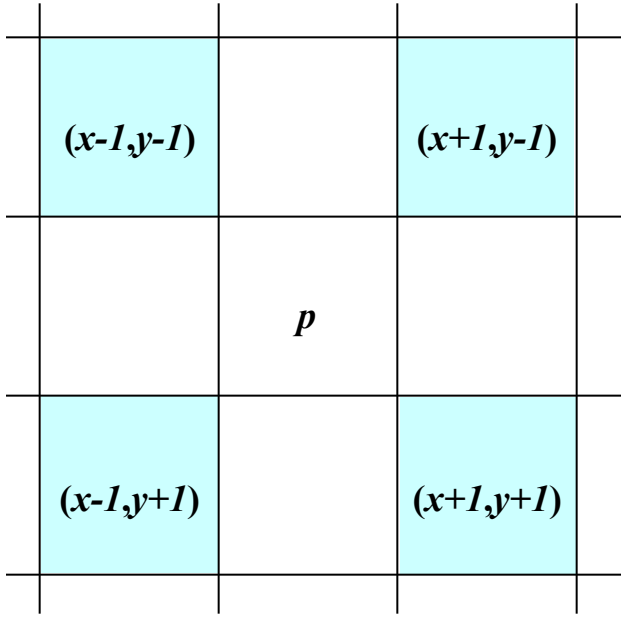
$(x-1,y-1)$	$(x,y-1)$	$(x+1,y-1)$
$(x-1,y)$	p	$(x+1,y)$
$(x-1,y+1)$	$(x,y+1)$	$(x+1,y+1)$

8-neighbors of *p*:

$$N_8(p) = \begin{Bmatrix} (x-1,y-1) \\ (x,y-1) \\ (x+1,y-1) \\ (x-1,y) \\ (x+1,y) \\ (x-1,y+1) \\ (x,y+1) \\ (x+1,y+1) \end{Bmatrix}$$

8-neighborhood relation considers all neighbor pixels.

Neighbors of a Pixel (cont.)



Diagonal neighbors of *p*:

$$N_D(p) = \begin{Bmatrix} (x-1,y-1) \\ (x+1,y-1) \\ (x-1,y+1) \\ (x+1,y+1) \end{Bmatrix}$$

Diagonal -neighborhood relation considers only diagonal neighbor pixels.

Connectivity

Connectivity is adapted from neighborhood relation. Two pixels are connected if they are in the same class (i.e. the same color or the same range of intensity) and they are neighbors of one another.

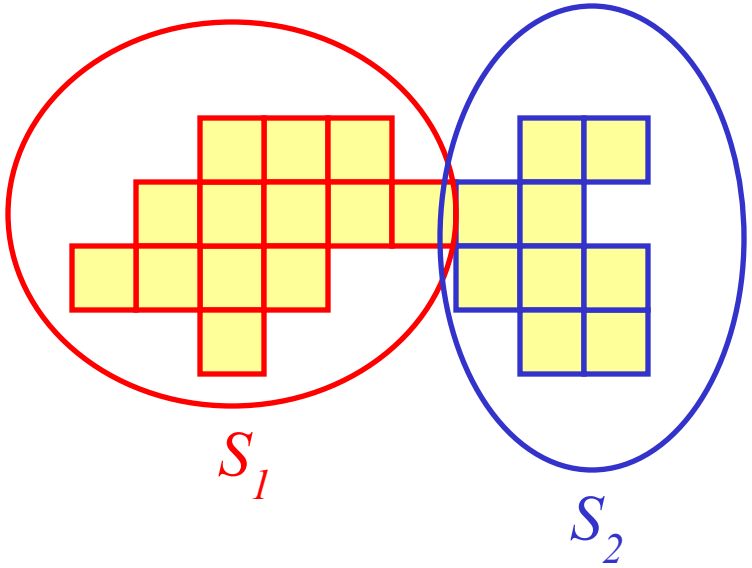
For p and q from the same class

- 4-connectivity: p and q are 4-connected if $q \in N_4(p)$
- 8-connectivity: p and q are 8-connected if $q \in N_8(p)$
- mixed-connectivity (m-connectivity):

  p and q are m-connected if $q \in N_4(p)$ or $q \in N_D(p)$ and $N_4(p) \cap N_4(q) = \varnothing$

Adjacency

A pixel p is *adjacent* to pixel q is they are connected. Two image subsets S_1 and S_2 are adjacent if some pixel in S_1 is adjacent to some pixel in S_2



We can define type of adjacency: 4-adjacency, 8-adjacency or m-adjacency depending on type of connectivity.

Path

A ***path*** from pixel p at (x,y) to pixel q at (s,t) is a sequence of distinct pixels:

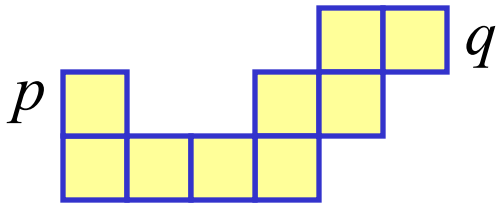
$$(x_0,y_0), (x_1,y_1), (x_2,y_2),\ldots, (x_n,y_n)$$

such that

$$(x_0,y_0) = (x,y) \text{ and } (x_n,y_n) = (s,t)$$

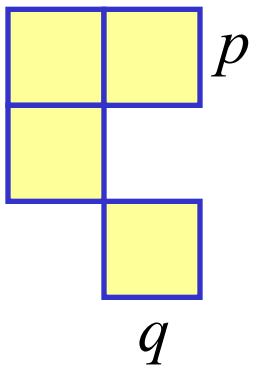
and

$$(x_i,y_i) \text{ is adjacent to } (x_{i-1},y_{i-1}), \qquad i = 1,\ldots,n$$

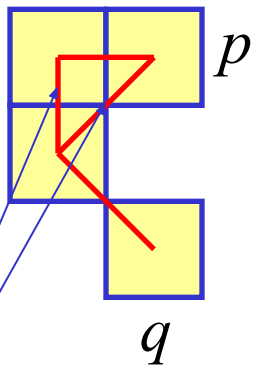


We can define type of path: 4-path, 8-path or m-path depending on type of adjacency.

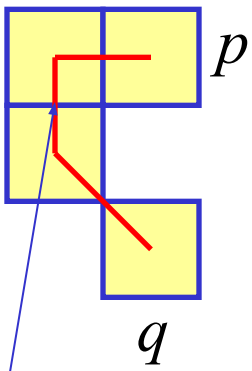
Path (cont.)



8-path



m-path



8-path from p to q
results in some ambiguity

m-path from p to q
solves this ambiguity

Distance

For pixel p, q, and z with coordinates (x,y), (s,t) and (u,v), D is a ***distance function*** or ***metric*** if

- $D(p,q) \geq 0 \quad (D(p,q) = 0$ if and only if $p = q)$
- $D(p,q) = D(q,p)$
- $D(p,z) \leq D(p,q) + D(q,z)$

Example: Euclidean distance

$$D_e(p,q) = \sqrt{(x-s)^2 + (y-t)^2}$$

Distance (cont.)

D_4-distance (*city-block distance*) is defined as

$$D_4(p,q) = |x-s| + |y-t|$$

		2		
	2	1	2	
2	1	0	1	2
	2	1	2	
		2		

Pixels with $D_4(p) = 1$ is 4-neighbors of p.

Distance (cont.)

D_8-distance (*chessboard distance*) is defined as

$$D_8(p,q) = \max(|x-s|,|y-t|)$$

2	2	2	2	2
2	1	1	1	2
2	1	0	1	2
2	1	1	1	2
2	2	2	2	2

Pixels with $D_8(p) = 1$ is 8-neighbors of p.